IEEE Recommended Practice for Establishing Liquid Immersed and Dry-Type Power and Distribution Transformer Capability when Supplying Nonsinusoidal Load Currents

IEEE Power and Energy Society

Sponsored by the
Transformers Committee

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(Revision of IEEE Std C57.110-2008)

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Abstract: Provided in this recommended practice are calculation methods for conservatively evaluating the feasibility for an existing installed dry-type or liquid immersed transformer, to supply nonsinusoidal load currents as a portion of the total load. Also provided is necessary application information to assist in properly specifying a new transformer expected to carry a load, a portion of which is composed of nonsinusoidal load currents. A number of examples illustrating these methods and calculations are presented. Reference annexes provide a comparison of the document calculations to calculations found in other industry standards. Suggested temperature rise calculation methods are detailed for reference purposes.

Keywords: current harmonics, harmonic loss factor, IEEE C57.110™, k-factor, nonsinusoidal load currents

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Introduction

This introduction is not part of IEEE Std C57.110-2018, IEEE Recommended Practice for Establishing Liquid Immersed and Dry-Type Power and Distribution Transformer Capability when Supplying Nonsinusoidal Load Currents.

One side effect of the trend toward energy efficiency is the increase in harmonic current flowing in the power system. The widespread use of solid state electronics in industrial and residential loads on small and medium power dry-type and liquid immersed transformers has resulted in a dramatic increase in the harmonic content of the load current of these transformers. It has become common for the harmonic factor of the current to exceed 0.05 per-unit, which is the limit specified for "usual service conditions" in IEEE Std C57.12.00™ and IEEE Std C57.12.01™. The higher harmonic content in the load current of these transformers causes higher eddy current loss in the windings and the structural parts linked by the transformer leakage flux and, consequently, higher operating temperatures. In addition, harmonic load currents reduce the efficiency of the transformer. Users of this document should also recognize that liquid immersed transformers may have different load limitations than dry-type transformers and that the harmonic loading practices should treat the two types of transformers differently when necessary.

This recommended practice provides guidance for the conservative loading of transformers carrying nonsinusoidal load currents so that overheating is avoided. The intent of this document is to provide simple methods of calculating these effects on either a new transformer or an existing transformer. More specifically, it is expected that this recommended practice would be used for the following situations:

a) A new transformer required to carry some nonsinusoidal load currents, but will not be entirely devoted to a rectifier load.

b) An existing transformer not originally specified for supplying nonsinusoidal load currents, but is now required to supply a load, a portion of which is nonsinusoidal.

Two methods are described in this recommended practice. The first method is intended to illustrate calculations by those with access to detailed information on loss density distribution within each of the transformer windings. The second method is less accurate and is intended for use by those with access to transformer certified test report data only. It is anticipated that the first method will emphasize the information necessary to specify a new transformer and show how this information is used by transformer design engineers, whereas the second method will be employed primarily by users. This recommended practice provides methods for conservatively evaluating the feasibility of applying nonsinusoidal load currents to existing transformers and clarifies the requirements for specifying new transformers to supply nonsinusoidal loads.

New transformers that are intended to supply loads with high harmonic content must be specified with a harmonic current spectrum. The designer cannot assume nor can the user expect the designer to use standard or typical current distribution tables. If the harmonic content of the load is unknown, then both the user and the transformer designer are at risk and reasonable steps should be taken to help ensure a conservative design for the application. Guidelines on how this information is used to develop proper transformer sizing is provided in this document, but appropriate calculations specific to the type of transformer design are the responsibility of the designer. Approximate calculation techniques that provide conservative results are provided for the typical user who has much less information than the transformer designer.

In the latest revision of this recommended practice, the document was updated to current IEEE styles, and general revisions were made. In addition, substantial additions were made to the references and three new annexes were added. Annex C provides information on skin effect in addition to eddy and other losses that were previously discussed. Annex F aids the user in understanding the impact of the highest eddy losses as compared to the average values, and Annex G provides sample loss data for eddy current and other stray losses across a range of distribution transformers.

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IEEE Recommended Practice for Establishing Liquid Immersed and Dry-Type Power and Distribution Transformer Capability when Supplying Nonsinusoidal Load Currents

1. Overview

1.1 Scope

This recommended practice applies only to two winding transformers covered by IEEE Std C57.12.00, IEEE Std C57.12.01, and NEMA ST20.[1] It does not apply to rectifier transformers.[2]

1.2 Purpose

The purpose of this document is to establish uniform methods for determining the capability of transformers when supplying nonsinusoidal load currents of known characteristics.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std C57.12.00™, IEEE Standard General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers.[3,4]

IEEE Std C57.12.01™, IEEE Standard General Requirements for Dry-Type Distribution and Power Transformers Including Those with Solid Cast and/or Resin-Encapsulated Windings.

[1]Information on references can be found in Clause 2.
[2]Rectifier transformers are addressed by IEEE Std C57.18.10™, IEEE Standard Practices and Requirements for Semiconductor Power Rectifier Transformers.
[3]IEEE publications are available from The Institute of Electrical and Electronics Engineers, 445 Hoes Lane, Piscataway, NJ 08854, USA (http://standards/ieee.org/).
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IEEE Std C57.12.90™, IEEE Standard Test Code for Liquid-Immersed Distribution, Power, and Regulating Transformers.

IEEE Std C57.12.91™, IEEE Standard Test Code for Dry-Type Distribution and Power Transformers.

IEEE Std C57.91™, IEEE Guide for Loading Mineral-Oil-Immersed Transformers.

IEEE Std C57.96™, IEEE Guide for Loading Dry-Type Distribution and Power Transformers.

IEEE Std C57.154™, IEEE Standard for the Design, Testing, and Application of Liquid-Immersed Distribution, Power, and Regulating Transformers Using High-Temperature Insulation Systems and Operating at Elevated Temperatures.

NEMA ST20, Dry-Type Transformers for General Applications.[5]

3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.[6]

harmonic loss factor: The ratio of the total winding eddy current losses due to the harmonics, to the winding eddy current losses at the power frequency, when no harmonic currents exist.

4. General considerations

4.1 Transformer losses

IEEE Std C57.12.90 and IEEE Std C57.12.91 categorize transformer losses as no-load loss (excitation loss), load loss (impedance loss), and total loss (the sum of no-load loss and load loss). Load loss is subdivided into I^2R loss and stray loss. Stray loss is determined by subtracting the I^2R loss (calculated from the measured resistance) from the measured load loss (impedance loss).

Stray loss can be defined as the loss due to stray electromagnetic flux in the windings, core, core clamps, magnetic shields, enclosure or tank walls, and so on. Thus, the stray loss is subdivided into winding stray loss and stray loss in components other than the windings (P_{OSL}). The winding stray loss includes winding conductor strand eddy-current loss and loss due to circulating currents between strands or parallel winding circuits. All of this loss may be considered to constitute winding eddy-current loss (P_{EC}). The total load loss can then be stated as follows in Equation (1):

$$P_{LL} = P + P_{EC} + P_{OSL} \tag{1}$$

where

P_{LL} is the load loss (watts)
P is the I^2R loss portion of the load loss (watts)
P_{EC} is the winding eddy-current loss (watts)
P_{OSL} is the other stray loss (watts)

[5]NEMA publications are available from Global Engineering Documents, 15 Inverness Way East, Englewood, CO 80112, USA (http://global.ihs.com/).
[6]*IEEE Standards Dictionary Online* is available at: http://dictionary.ieee.org.

4.1.1 Harmonic current effect on I^2R loss

If the rms value of the load current is increased due to harmonic components, the I^2R loss will be increased accordingly.

4.1.2 Harmonic current effect on winding eddy-current loss

Winding eddy-current loss (P_{EC}) in the frequency range under consideration (power frequency and associated harmonics) tends to be proportional to the square of the load current and approximately proportional to the square of frequency (see Crepaz [B8], Blume et al. [B6], Dwight [B11], as well as Bishop and Gilker [B5]).[7] It is this characteristic that can cause excessive winding loss and hence abnormal winding temperature rise and hottest spot temperatures in transformers supplying nonsinusoidal load currents.

4.1.3 Harmonic current effect on other stray loss

It is recognized that other stray loss (P_{OSL}) in the core, clamps, and structural parts will also increase at a rate proportional to the square of the load current. However, these losses will not increase at a rate proportional to the square of the frequency, as in the winding eddy losses. Studies by manufacturers and other researchers have shown that the eddy-current losses in bus bars, connections, and structural parts increase by a harmonic exponent factor of 0.8 or less. Therefore, as a conservative estimate, an exponent of 0.8 will be used throughout this document.[8] The effects of these losses will also vary depending on the type of transformer. For example, the temperature rise in these non-winding parts will generally not be very critical for dry-type transformers. However, these losses must be properly accounted for in liquid immersed transformers, since they contribute to additional heating of the insulating liquid and the hottest spots in the structural parts.

4.1.4 DC components of load current

Harmonic load currents are frequently accompanied by a dc component in the load current. A small dc component of load current will increase the transformer core loss slightly, but it will increase the magnetizing current and audible sound level more substantially. Relatively small dc components (up to the rms magnitude of the transformer excitation current at rated voltage) are expected to have no effect on the load carrying capability of a transformer determined by this recommended practice. Higher dc load current components may adversely affect transformer capability; possibly causing core saturation, core heating and unusual leakage flux patterns accompanied by unusual stray loss effects and should be avoided or otherwise accounted for.

4.1.5 Effect on top-liquid rise

For liquid immersed transformers, the top-liquid rise (θ_{TO}) will increase as the total load losses increase due to harmonic loading. Unlike dry-type transformers, the other stray loss (P_{OSL}) must be considered, since these losses also affect the top-liquid rise.

4.2 Transformer capability equivalent

The transformer capability established by following the procedures in this recommended practice is based on the following premises:

a) The transformer, except for the load harmonic current spectrum, is presumed to be operated in accordance with Usual Service Conditions in IEEE Std C57.12.00 or IEEE Std C57.12.01..

[7] The numbers in brackets correspond to those of the bibliography in Annex A.
[8] See paragraph 2 of Annex B for additional information on this topic.

b) The transformer is presumed to be capable of supplying a load current of any harmonic content provided that the total load loss, the load loss in each winding, and the loss density in the region of the highest eddy-current loss do not exceed the levels for full load, rated frequency, and sine wave design conditions. It is also presumed that the limiting condition is the loss density in the region of highest winding eddy-current loss; hence, this is the basis used for establishing capability equivalency.[9]

4.3 Basic data

In order to perform the calculations in this recommended practice, the characteristics of the nonsinusoidal load current must be defined either in terms of the magnitude of the fundamental frequency component or the magnitude of the total rms current. Each harmonic frequency component must also be defined from power system measurements. In addition, information on the magnitude of winding eddy-current loss density must be available.

4.4 Transformer per-unit losses

Since the greatest concern about a transformer operating under harmonic load conditions will be for overheating of the windings, it is convenient to consider loss density in the windings on a per-unit basis (base current is rated current and base loss density is the I^2R loss density at rated current). Thus, Equation (1) applied to rated load conditions can be rewritten on a per-unit basis as follows in Equation (2):

$$P_{LL-R}(\mathrm{pu}) = 1 + P_{EC-R}(\mathrm{pu}) + P_{OSL-R}(\mathrm{pu}) \tag{2}$$

where

$P_{LL-R}(\mathrm{pu})$ is the per-unit load loss under rated conditions
$P_{EC-R}(\mathrm{pu})$ is the per-unit winding eddy-current loss under rated conditions
$P_{OSL-R}(\mathrm{pu})$ is the per-unit other stray loss under rated conditions

Given the eddy-current loss under rated conditions for a transformer winding or portion of a winding, $\left(P_{EC-R}\right)$, the eddy-current loss due to any defined nonsinusoidal load current can be expressed as follows in Equation (3):

$$P_{EC} = P_{EC-R} \sum_{h=1}^{h=h_{\max}} \left(\frac{I_h}{I_R}\right)^2 h^2 \tag{3}$$

where

P_{EC} is the winding eddy-current loss (watts)
$P_{EC-R}(\mathrm{pu})$ is the winding eddy-current loss under rated conditions (watts)
h is the harmonic order
$h_{\max}$ is the highest significant harmonic number
I_h is the rms current at harmonic h (amperes)
I_R is the rms fundamental current under rated frequency and rated load conditions (amperes)

[9]The simple methods of calculation of transformer capability equivalent given in this document neglect the skin effect, which becomes more pronounced at frequencies in the upper portion of the frequency range under consideration. See Annex C for a more rigorous analysis that includes the skin effect and compares the results of the two calculation methods.

The I^2R loss at rated load is one per-unit (by definition). For nonsinusoidal load currents, the equation for the rms current in per-unit form (base current is rated current), will be as follows in Equation (4):

$$I(\mathrm{pu}) = \sqrt{\sum_{h=1}^{h=h_{\max}} I_h^2(\mathrm{pu})} \tag{4}$$

where

$I(\mathrm{pu})$ is the per-unit rms load current
h is the harmonic order
$h_{\max}$ is the highest significant harmonic number
$I_h(\mathrm{pu})$ is the per-unit rms current at harmonic h

Equation (3) can also be written in per-unit form (base current is the rated current and base loss density is the I^2R loss density at rated current) as follows in Equation (5):

$$P_{EC}(\mathrm{pu}) = P_{EC-R}(\mathrm{pu}) \sum_{h=1}^{h=h_{\max}} I_h^2(\mathrm{pu}) h^2 \tag{5}$$

where

$P_{EC}(\mathrm{pu})$ is the per-unit winding eddy-current loss
$P_{EC-R}(\mathrm{pu})$ is the per-unit winding eddy-current loss under rated conditions
h is the harmonic order
$h_{\max}$ is the highest significant harmonic number
$I_h(\mathrm{pu})$ is the per-unit rms current at harmonic h

4.5 Transformer losses at measured currents

Equation (2) through Equation (5) assume that the measured winding currents are taken at the rated currents of the transformer. Since this is seldom encountered in the field, a new term is needed to describe the winding eddy losses at the measured current and the power frequency $P_{\text{EC-O}}$. Three assumptions in addition to the basic premises of the Transformer Capability Equivalent are necessary to clarify the use of this term:

a) The eddy losses are approximately proportional to the square of the frequency. This assumption will cause any subsequent equations to be accurate for small conductors and low harmonics, providing a conservative calculation, for a combination of larger conductors and higher harmonics.

b) The eddy losses are a function of the current in the conductors. Any equation for loss can then be expressed in terms of the rms load current I.

c) Superposition of eddy losses will apply, which will permit the direct addition of eddy losses due to the various harmonics.

Equation (3) and Equation (5) may now be written more generally as in Equation (6):

$$P_{EC} = P_{EC-O} \sum_{h=1}^{h=h_{\max}} \left(\frac{I_h}{I}\right)^2 h^2 \tag{6}$$

where

P_{EC}	is the winding eddy-current loss (watts)
$P_{\mathrm{EC-O}}$	is the winding eddy-current loss at the measured current and the power frequency (watts)
h	is the harmonic order
$h_{\max}$	is the highest significant harmonic number
I_h	is the rms current at harmonic h (amperes)
I	is the rms load current (amperes)

By removing the rms current, I from the summation, we now have Equation (7):

$$P_{EC} = P_{EC-O} \times \frac{\sum_{h=1}^{h=h_{\max}} I_h^{\,2} h^2}{I^2} \tag{7}$$

where

P_{EC}	is the winding eddy-current loss (watts)
$P_{\mathrm{EC-O}}$	is the winding eddy-current loss at the measured current and the power frequency (watts)
h	is the harmonic order
$h_{\max}$	is the highest significant harmonic number
I_h	is the rms current at harmonic h (amperes)
I	is the rms load current (amperes)

The rms value of the nonsinusoidal load current is then given by Equation (8):

$$I = \sqrt{\sum_{h=1}^{h=h_{\max}} I_h^{\,2}} \tag{8}$$

where

I	is the rms load current (amperes)
h	is the harmonic order
$h_{\max}$	is the highest significant harmonic number
I_h	is the rms current at harmonic h (amperes)

The rms current, I may be expressed in terms of the component frequencies as follows in Equation (9):

$$P_{EC} = P_{EC-O} \times \frac{\sum_{h=1}^{h=h_{\max}} I_h^{\,2} h^2}{\sum_{h=1}^{h=h_{\max}} I_h^{\,2}} \tag{9}$$

where

P_{EC}	is the winding eddy-current loss (watts)
$P_{\mathrm{EC-O}}$	is the winding eddy-current loss at the measured current and the power frequency (watts)
h	is the harmonic order
$h_{\max}$	is the highest significant harmonic number
I_h	is the rms current at harmonic h (amperes)

4.6 Harmonic loss factor[10] for winding eddy currents

It is convenient to define a single number that may be used to determine the capabilities of a transformer in supplying power to a load. F_{HL} is a proportionality factor applied to the winding eddy losses, which represents the effective rms heating as a result of the harmonic load current. F_{HL} is the ratio of the total winding eddy current losses due to the harmonics, P_{EC}, to the winding eddy current losses at the power frequency, when no harmonic currents exist (P_{EC-O}). This definition in equation form is as follows in Equation (10):

$$F_{HL} = \frac{P_{EC}}{P_{EC-O}} = \frac{\sum_{h=1}^{h=h_{max}} I_h^{\,2} h^2}{\sum_{h=1}^{h=h_{max}} I_h^{\,2}} \tag{10}$$

where

F_{HL}	is the harmonic loss factor for winding eddy currents
P_{EC}	is the winding eddy-current loss (watts)
P_{EC-O}	is the winding eddy-current loss at the measured current and the power frequency (watts)
h	is the harmonic order
h_{max}	is the highest significant harmonic number
I_h	is the rms current at harmonic h (amperes)

Equation (10) permits F_{HL} to be calculated in terms of the actual rms values of the harmonic currents. Harmonic analyzers permit calculations to be made in terms of the harmonics normalized to the total rms current or to the first or fundamental harmonic. Equation (10) may be adapted to these situations by dividing the numerator and denominator by either I, the rms load current, or by I_1, the rms fundamental load current. These terms may now be applied to Equation (10) term by term, resulting in Equation (11) and Equation (12):

$$F_{HL} = \frac{\sum_{h=1}^{h=h_{max}} \left[\frac{I_h}{I}\right]^2 h^2}{\sum_{h=1}^{h=h_{max}} \left[\frac{I_h}{I}\right]^2} \tag{11}$$

where

F_{HL}	is the harmonic loss factor for winding eddy currents
h	is the harmonic order
h_{max}	is the highest significant harmonic number
I_h	is the rms current at harmonic h (amperes)
I	is the rms load current (amperes)

$$F_{HL} = \frac{\sum_{h=1}^{h=h_{max}} \left[\frac{I_h}{I_1}\right]^2 h^2}{\sum_{h=1}^{h=h_{max}} \left[\frac{I_h}{I_1}\right]^2} \tag{12}$$

[10]The harmonic loss factor is similar but not identical to the K-factor referenced in other standards. For a comparison of the harmonic loss factor with the K-factor definition referenced in UL standards, see Annex D.

where

F_{HL} is the harmonic loss factor for winding eddy currents
h is the harmonic order
h_{max} is the highest significant harmonic number
I_h is the rms current at harmonic h (amperes)
I_1 is the rms fundamental load current (amperes)

Note that the quantity I_h / I_1 may be directly read on a meter, bypassing the computation procedure.

In either case, F_{HL} remains the same value, since it is a function of the harmonic current spectrum and is independent of the relative magnitude. Two examples may be used to clarify these definitions. In both examples, a nonsinusoidal load current of 1804 A rms will be used as the rated current. The load may be described by the harmonic distribution in Table 1, normalized to the rms load current of 1804 A.

Table 1—Example of a harmonic distribution normalized to the rms load current

h	I_h	$\frac{I_h}{I}$
1	1764	0.97783
5	309	0.17129
7	195	0.10809
11	79.4	0.04401
13	50.5	0.02799
17	27.1	0.01502
19	17.7	0.00981

The calculation is tabulated in Table 2.

Table 2—Tabulated calculation of Table 1 harmonic distribution

h	$\frac{I_h}{I}$	$\left(\frac{I_h}{I}\right)^2$	h^2	$\left(\frac{I_h}{I}\right)^2 h^2$
1	0.97783	0.95615	1	0.95615
5	0.17129	0.029339	25	0.73347
7	0.10809	0.011684	49	0.57252
11	0.04401	0.0019372	121	0.23440
13	0.02799	0.00078363	169	0.13243
17	0.01502	0.00022567	289	0.065217
19	0.00981	0.000096266	361	0.034752
Σ		1.000		2.729

The summation of the third column $(I_h / I)^2$ is equal to 1.00 and represents the rated rms load on a per-unit basis. The harmonic loss factor for this harmonic distribution, using Equation (11) is as follows:

$$F_{HL} = \frac{\sum_{h=1}^{h=h_{max}} \left[\frac{I_h}{I}\right]^2 h^2}{\sum_{h=1}^{h=h_{max}} \left[\frac{I_h}{I}\right]^2} = \frac{2.729}{1.000} = 2.73$$

where

F_{HL} is the harmonic loss factor for winding eddy currents
h is the harmonic order
h_{max} is the highest significant harmonic number
I_h is the rms current at harmonic h (amperes)
I is the rms load current (amperes)

This same loading example may also be described in terms of the harmonic currents normalized to the harmonic current of the rms fundamental frequency, as shown in Table 3.

Table 3—Example of a harmonic distribution normalized to the rms fundamental load current

h	I_h	$\frac{I_h}{I_1}$
1	1764	1.0000
5	309	0.17517
7	195	0.11054
11	79.4	0.045011
13	50.5	0.028628
17	27.1	0.015363
19	17.7	0.010034

Note that the values of the harmonic current I_h are the same in both examples, but the normalized values are different since these values are normalized to the harmonic current of the fundamental frequency. The calculation is tabulated in Table 4.

Table 4—Tabulated calculation of Table 3 harmonic distribution

h	$\frac{I_h}{I_1}$	$\left(\frac{I_h}{I_1}\right)^2$	h^2	$\left(\frac{I_h}{I_1}\right)^2 h^2$
1	1.0000	1.0000	1	1.0000
5	0.17517	0.030685	25	0.76711
7	0.11054	0.012219	49	0.59874
11	0.04501	0.0020260	121	0.24514
13	0.028628	0.00081956	169	0.13851
17	0.015363	0.00023602	289	0.068210
19	0.010034	0.00010068	361	0.036346
Σ		1.046		2.854

The summation of the third column $(I_h / I_1)^2$ is equal to 1.05 and represents the rated rms fundamental load on a per-unit basis. The harmonic loss factor for this harmonic distribution using Equation (12) is as follows:

$$F_{HL} = \frac{\sum_{h=1}^{h=h_{\max}} \left[\frac{I_h}{I_1}\right]^2 h^2}{\sum_{h=1}^{h=h_{\max}} \left[\frac{I_h}{I_1}\right]^2} = \frac{2.854}{1.046} = 2.73$$

where

F_{HL} is the harmonic loss factor for winding eddy currents
h is the harmonic order
$h_{\max}$ is the highest significant harmonic number
I_h is the rms current at harmonic h (amperes)
I_1 is the rms fundamental load current (amperes)

Whether the individual harmonic currents are normalized to the rms load current I or to the rms fundamental load current I_1, the value of harmonic loss factor is the same:

$$F_{HL} = \frac{2.729}{1.000} = \frac{2.854}{1.046} = 2.73$$

4.7 Harmonic loss factor for other stray losses

The heating due to other stray losses is generally not a consideration for dry-type transformers, since the heat generated is dissipated by the cooling air. However, these losses can have a substantial effect on liquid immersed transformers, by causing additional heating of the cooling liquid. A relationship similar to the harmonic loss factor for winding eddy losses exists for these other stray losses in a transformer, and may be developed in a similar manner. However, the losses due to bus bar connections, structural parts, tank, and so on, are proportional to the square of the load current and the harmonic frequency to the 0.8 power, as stated in 4.1.3. This may be expressed in a form similar to Equation (3), as shown in Equation (13):

$$P_{OSL} = P_{OSL-R} \sum_{h=1}^{h=h_{\max}} \left(\frac{I_h}{I_R}\right)^2 h^{0.8} \qquad (13)$$

where

P_{OSL} is the other stray loss (watts)
$P_{\mathrm{OSL-R}}$ is the other stray loss under rated conditions (watts)
h is the harmonic order
$h_{\max}$ is the highest significant harmonic number
I_h is the rms current at harmonic h (amperes)
I_{R} is the rms fundamental current under rated frequency and rated load conditions (amperes)

The equations corresponding to the harmonic loss factor, normalized to the rms current and normalized to the rms fundamental current, respectively, are as follows in Equation (14) and Equation (15):

$$F_{HL-STR} = \frac{\sum_{h=1}^{h=h_{\max}} \left[\frac{I_h}{I}\right]^2 h^{0.8}}{\sum_{h=1}^{h=h_{\max}} \left[\frac{I_h}{I}\right]^2} \tag{14}$$

where

$F_{\mathrm{HL-STR}}$ is the harmonic loss factor for other stray losses
h is the harmonic order
$h_{\max}$ is the highest significant harmonic number
I_h is the rms current at harmonic h (amperes)
I is the rms load current (amperes)

$$F_{HL-STR} = \frac{\sum_{h=1}^{h=h_{\max}} \left[\frac{I_h}{I_1}\right]^2 h^{0.8}}{\sum_{h=1}^{h=h_{\max}} \left[\frac{I_h}{I_1}\right]^2} \tag{15}$$

where

$F_{\mathrm{HL-STR}}$ is the harmonic loss factor for other stray losses
h is the harmonic order
$h_{\max}$ is the highest significant harmonic number
I_h is the rms current at harmonic h (amperes)
I_1 is the rms fundamental load current (amperes)

Using the harmonic distribution from the last example of 4.6, the calculation is tabulated for the normalized fundamental current base, as shown in Table 5.

Table 5—Tabulated calculation of the harmonic loss factor for other stray losses

h	$\frac{I_h}{I_1}$	$\left(\frac{I_h}{I_1}\right)^2$	$h^{0.8}$	$\left(\frac{I_h}{I_1}\right)^2 h^{0.8}$
1	1.0000	1.0000	1.0000	1.0000
5	0.17517	0.030685	3.6239	0.11120
7	0.11054	0.012219	4.7433	0.057959
11	0.045010	0.0020259	6.8095	0.013795
13	0.028628	0.00081956	7.7831	0.0063788
17	0.015363	0.00023602	9.6463	0.0022767
19	0.010034	0.00010068	10.544	0.0010616
Σ		1.046		1.193

The fifth column summation divided by the third column summation results in a harmonic loss factor for other stray losses:

$$F_{HL-STR} = \frac{\sum_{h=1}^{h=h_{\max}} \left[\frac{I_h}{I_1}\right]^2 h^{0.8}}{\sum_{h=1}^{h=h_{\max}} \left[\frac{I_h}{I_1}\right]^2} = \frac{1.193}{1.046} = 1.14$$

5. Design considerations for new transformer specification

5.1 Harmonic current filtering

When it is practical, the user may install filters on the secondary line to reduce some of the harmonic load currents supplied to the transformer. However, caution should be exercised, since current amplification at this frequency may occur, if one of the harmonic frequencies is close to the resonant frequency resulting from the filtering circuit.

5.2 Impact on the neutral

Zero sequence currents flow in the neutral when the harmonic current frequencies include harmonic orders having multiples of three (3, 6, 9, etc.). Oversizing of this neutral may be required. In those circumstances, a common practice with low-voltage general-purpose transformers is to double the neutral ampacity.

5.3 Power factor correction equipment

Power factor correction equipment is frequently installed to decrease utility costs. Care should be taken when this is done, since current amplification at certain frequencies due to resonance in the circuit can be quite high. In addition, the inductance that is reduced in the circuit generally allows higher harmonic currents to exist in the system. Harmonic heating effects from these conditions may be damaging to transformers and other equipment. The additional losses produced may also increase utility costs due to increased wattage requirements, even though the load power factor was improved.

5.4 Electrostatic ground shields

Electrostatic ground shields are frequently specified between the primary and secondary windings. The presence of electrostatic ground shields tends to reduce capacitive coupling between the windings. This reduces the coupling of transients between the two windings. Line disturbances produced by converter equipment connected to the transformer secondary will be reduced, but will not be eliminated on the primary side of the transformer. The shields are not intended to reduce harmonic currents, but by virtue of their magnetic coupling to windings carrying such currents, additional heating losses are induced. The shields have only a small effect on harmonic currents. Harmonic current filtering is necessary to obtain significant reductions in harmonic current content. Since the shields are in the high magnetic field between the primary and secondary windings, eddy current losses occur in the shield that are increased by harmonic currents just as the winding eddy current losses are increased by the harmonic currents. It is important that the shields are designed so that the eddy losses in the shields produced by the harmonic current content do not result in an excessive temperature rise in the shield.

The electrostatic shields also serve as protection to the secondary side of the transformer from transients that may be impressed on the high-voltage winding. This is especially important for transformers with ungrounded secondary windings. Transients on the high-voltage side of a transformer can dramatically increase the surge voltage seen on an ungrounded secondary winding from what may have been expected for a grounded winding. This may damage transformer windings and parts or equipment connected on the secondary side of the transformer. The presence of an electrostatic ground shield between the primary and secondary windings reduces the magnitude of the transient coupled to the secondary windings.

5.5 Design consideration outside the windings

Harmonic currents can substantially increase the stray losses in structural parts outside of the windings. Additional clearances, the use of nonmagnetic materials in place of mild steel, the break up of potential circulation current paths, and the use of shielding materials should all be considered as ways to reduce the effects of harmonic current heating in structural parts. These other stray losses P_{OSL} must be included in the losses used to determine top-liquid rise $\Delta\theta_{\mathrm{TO}}$ under harmonic loading conditions.

5.6 Harmonic spectrum analysis

It is preferred that the harmonic spectrum to which the transformer will be subjected be specified to the transformer manufacturer at the time of inquiry. An accurate analysis for proper sizing of the transformer can only be made by evaluating the specific harmonic spectrum. The harmonic spectrum supplied should be identified as to whether it is measured on the primary or secondary side of the transformer. If the harmonic spectrum is provided in per-unit form, then the fundamental should be defined at the rated frequency of the transformer specified. If the spectrum cannot be supplied, then the user's calculation or estimate of F_{HL} should be specified. However, in this case F_{HL} should be conservatively estimated by the user to compensate for the lack of specific loading information. The specifying engineer must supply this loading information, since the transformer manufacturer cannot assume values with no real knowledge of the system to which the transformer will be applied.

5.7 Design consideration in the windings

Since harmonic currents can substantially increase the eddy-current losses in the windings, this increase of losses must be considered in the temperature rise calculation when a new transformer is specified. For each winding, the per-unit eddy-current losses in the region of highest loss density can be defined for rated frequency operation at rated current by the transformer manufacturer in terms of Equation (2), with $P_{\mathrm{OSL}\text{-}R}(\mathrm{pu})$ equal to zero (since there is no other stray loss in the windings by definition). The per-unit loss density in these regions of highest eddy-current loss can then be recalculated for the defined nonsinusoidal load current by combining Equation (2), Equation (5), Equation (8), and Equation (11), as shown in Equation (16):

$$P_{LL}\left(\mathrm{pu}\right) = I^2\left(\mathrm{pu}\right) \times \left(1 + F_{HL} \times P_{EC-R}\left(\mathrm{pu}\right)\right) \tag{16}$$

where

$P_{LL}\left(\mathrm{pu}\right)$ is the per-unit load loss
$I\left(\mathrm{pu}\right)$ is the per-unit rms current
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{EC-R}\left(\mathrm{pu}\right)$ is the per-unit winding eddy-current loss under rated conditions

To adjust the per-unit loss density in the individual windings, the effect of F_{HL} must be known on each winding. Thus, the low-voltage winding, with its larger conductor cross section may start with a lower loss density and a lower temperature rise, but it may increase more than the high-voltage winding and exhibit the hottest spot in the transformer for harmonic loads. That is to say, there is one value of F_{HL} for the load, but the effects on different transformers and different windings within the same transformer can be different. For liquid immersed transformers, heating of the liquid by stray losses other than the eddy losses also affects the temperature rise of the windings.

In these mentioned regions, considering the per-unit loss density obtained by Equation (16) with a nonsinusoidal load current of 1.0 (pu) rms magnitude, the limits of temperature and temperature rise given in IEEE Std C57.12.01 and IEEE Std C57.12.00 and IEEE Std C57.154 must be met. See Annex E for suggested procedures for performing a temperature rise test to validate the thermal performance of a design.

6. Recommended procedures for evaluating the load capability of existing transformers[11]

6.1 Transformer capability equivalent calculation using design eddy-current loss data

6.1.1 Typical calculations for dry-type transformers

The per-unit eddy-current loss in the region of highest loss density can be defined for rated frequency operation at rated current by the transformer manufacturer in terms of Equation (2), with $P_{\mathrm{OSL\text{-}R}}(\mathrm{pu})$ equal to zero (since there is no other stray loss in the windings by definition).

The per-unit value of nonsinusoidal load current that will make the result of the Equation (16) calculation equal to the design value of loss density in the highest loss region for rated frequency and for rated current operation is given by Equation (17). This assumes that the normal life of the unit will be maintained. However, it is permissible to overload a unit with a resulting loss of life, as described in the loading guide, IEEE Std C57.96.

$$I_{\max}(\mathrm{pu}) = \sqrt{\frac{P_{LL-R}(\mathrm{pu})}{1+F_{HL}\times P_{EC-R}(\mathrm{pu})}} \tag{17}$$

where

$I_{\max}(\mathrm{pu})$	is the max permissible rms nonsinusoidal load current under rated conditions
$P_{LL-R}(\mathrm{pu})$	is the per-unit load loss under rated conditions
F_{HL}	is the harmonic loss factor for winding eddy currents
$P_{EC-R}(\mathrm{pu})$	is the per-unit winding eddy-current loss under rated conditions

Two examples illustrate the use of these formulas.

6.1.1.1 Example 1

Given a nonsinusoidal load current with the following harmonic distribution, determine the maximum load current that can be continuously drawn (under standard conditions) from an IEEE standard transformer having a rated full load current of 1200 A and whose winding eddy-current loss under rated conditions ($P_{\mathrm{EC\text{-}R}}$) at the point of maximum loss density is 15% of the local I^2R loss (see Table 6).

Table 6—Harmonic distribution for maximum load current Example 1

h	1	5	7	11	13	17	19
$\frac{I_h}{I_1}$	1.00	0.23	0.11	0.042	0.027	0.013	0.0080

The maximum per-unit local loss density under rated conditions $P_{\mathrm{LL\text{-}R}}(\mathrm{pu})$ is then 1.15 pu. Equation (16) and Equation (17) require values for $I_h{}^2(\mathrm{pu})$, h^2, and $I_h{}^2(\mathrm{pu})\,h^2$. These can be calculated and tabulated as in Table 7.

[11]The user is cautioned that local and national electrical codes should be consulted before any installed unit is officially de-rated, such as changing the nameplate. Some units may not be de-rated without violating these codes.

Table 7—Tabulated calculation of the harmonic loss factor for Example 1

h	$\dfrac{I_h}{I_1}$	$\left(\dfrac{I_h}{I_1}\right)^2$	h^2	$\left(\dfrac{I_h}{I_1}\right)^2 h^2$
1	1.00	1.0000	1	1.0000
5	0.23	0.052900	25	1.3225
7	0.11	0.012100	49	0.59290
11	0.042	0.0017640	121	0.21344
13	0.027	0.00072900	169	0.12320
17	0.013	0.00016900	289	0.048841
19	0.0080	0.000064000	361	0.023104
$\sum$		1.068		3.324

The fifth column summation divided by the third column summation results in a harmonic loss factor of 3.112. From Equation (16), the local loss density for the nonsinusoidal load current is as follows:

$$P_{LL}(\text{pu}) = I^2(\text{pu}) \times \left(1 + F_{HL} \times P_{EC-R}(\text{pu})\right) = 1.068 \times (1 + 3.112 \times 0.150) = 1.57$$

where

$P_{LL}(\text{pu})$ is the per-unit load loss
$I(\text{pu})$ is the per-unit rms current
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{EC-R}(\text{pu})$ is the per-unit winding eddy-current loss under rated conditions

The maximum permissible per-unit nonsinusoidal load current with the given harmonic composition, from Equation (17), is as follows:

$$I_{\max}(\text{pu}) = \sqrt{\frac{P_{LL-R}(\text{pu})}{1 + F_{HL} \times P_{EC-R}(\text{pu})}} = \sqrt{\frac{1.150}{1 + (3.112 \times 0.150)}} = 0.885$$

where

$I_{\max}(\text{pu})$ is the max permissible rms nonsinusoidal load current under rated conditions
$P_{LL-R}(\text{pu})$ is the per-unit load loss under rated conditions
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{EC-R}(\text{pu})$ is the per-unit winding eddy-current loss under rated conditions

The maximum permissible per-unit nonsinusoidal load current with the given harmonic composition is as follows:

$$I_{\max} = 0.885 \times 1200 = 1062\ \text{A}$$

Thus, with the given nonsinusoidal load current harmonic composition, the transformer capability is approximately 89% of its sinusoidal load current capability, or 1062 A.

6.1.1.2 Example 2

Table 8 provides an example of a nonsinusoidal load current that has a strong third harmonic content with a harmonic distribution.

Table 8—Harmonic distribution for maximum load current Example 2

h	1	3	5	7	9	11	13	15	17	19
$\frac{I_h}{I}$	0.97	0.37	0.35	0.10	0.028	0.11	0.071	0.026	0.057	0.047

Determine the maximum load current that can be continuously drawn (under standard conditions) from a 225 kVA standard transformer having a rated full load secondary current of 624.5 A and whose average winding eddy-current loss under rated conditions ($P_{\text{EC-R}}$) at the point of maximum loss density is 12% of the local I^2R loss.

The maximum per-unit local loss density under rated conditions $P_{\text{LL-R}}$(pu) is then 1.12 pu. Equation (16) and Equation (17) require values for I_h^2(pu), h^2, and I_h^2(pu)h^2. These equations can be calculated and tabulated as in Table 9.

Table 9—Tabulated calculation of the harmonic loss factor for Example 2

h	$\frac{I_h}{I}$	$\left(\frac{I_h}{I}\right)^2$	h^2	$\left(\frac{I_h}{I}\right)^2 h^2$
1	0.97	0.94090	1	0.94090
3	0.37	0.13690	9	1.2321
5	0.35	0.12250	25	3.0625
7	0.10	0.010000	49	0.49000
9	0.028	0.00078400	81	0.063504
11	0.11	0.012100	121	1.4641
13	0.071	0.0050410	169	0.85193
15	0.026	0.00067600	225	0.15210
17	0.057	0.0032490	289	0.93896
19	0.047	0.0022090	361	0.79745
Σ		1.234		9.994

The fifth column summation divided by the third column summation results in a harmonic loss factor of 8.10. From Equation (16), the local loss density for the nonsinusoidal load current is as follows:

$$P_{LL}(\text{pu}) = I^2(\text{pu}) \times \left(1 + F_{HL} \times P_{EC-R}(\text{pu})\right) = 1.234 \times (1 + 8.099 \times 0.120) = 2.43$$

where

$P_{LL}(\text{pu})$ is the per-unit load loss
$I(\text{pu})$ is the per-unit rms current
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{EC-R}(\text{pu})$ is the per-unit winding eddy-current loss under rated conditions

and the maximum permissible nonsinusoidal load current with the given harmonic composition, from Equation (17), is as follows:

$$I_{\max}(\mathrm{pu}) = \sqrt{\frac{P_{LL-R}(\mathrm{pu})}{1+F_{HL}\times P_{EC-R}(\mathrm{pu})}} = \sqrt{\frac{1.120}{1+(8.099\times 0.120)}} = 0.754$$

where

$I_{\max}(\mathrm{pu})$	is the max permissible rms nonsinusoidal load current under rated conditions
$P_{LL-R}(\mathrm{pu})$	is the per-unit load loss under rated conditions
F_{HL}	is the harmonic loss factor for winding eddy currents
$P_{EC-R}(\mathrm{pu})$	is the per-unit winding eddy-current loss under rated conditions

The maximum permissible per-unit nonsinusoidal load current with the given harmonic composition is as follows:

$$I_{\max} = 0.754\times 624.5 = 471\ \mathrm{A}$$

Therefore, with the given nonsinusoidal load current harmonic composition, the transformer capability is approximately 75% of its sinusoidal load current capability, or 471 A.

6.1.2 Typical calculations for liquid immersed transformers

The calculations for liquid immersed transformers are similar to the dry-type transformers, except the effect of all stray losses must be addressed. As indicated by equations in IEEE Std C57.91, for self-cooled ONAN mode, the top-liquid rise is proportional to the total losses to the 0.8 exponent and may be estimated for the harmonic losses, based on rated load and losses, as shown in Equation (18) and Equation (19):

$$\theta_{TO} = \theta_{TO-R}\times\left(\frac{P_{LL}+P_{NL}}{P_{LL-R}+P_{NL}}\right)^{0.8} \tag{18}$$

where

θ_{TO}	is the top-liquid-rise over ambient temperature (°C)
$\theta_{\mathrm{TO-R}}$	is the top-liquid-rise over ambient temperature under rated conditions (°C)
P_{LL}	is the load loss (watts)
P_{LL-R}	is the load loss under rated conditions (watts)
P_{NL}	is the no load loss (watts)

$$P_{LL} = P + F_{HL}\times P_{EC} + F_{HL-STR}\times P_{OSL} \tag{19}$$

where

P_{LL}	is the load loss (watts)
P	is the I^2R loss portion of the load loss (watts)
F_{HL}	is the harmonic loss factor for winding eddy currents
P_{EC}	is the winding eddy-current loss (watts)
F_{HL-STR}	is the harmonic loss factor for other stray losses
P_{OSL}	is the other stray loss (watts)

The winding hottest spot conductor rise is also proportional to the load losses to the 0.8 exponent and may be calculated as in Equation (20):

$$\theta_g = \theta_{g-R} \times \left(\frac{P_{LL}\,(\text{pu})}{P_{LL-R}\,(\text{pu})} \right)^{0.8} \tag{20}$$

where

θ_{g}	is the hottest-spot conductor rise over top-liquid temperature (°C)
$\theta_{\mathrm{g-R}}$	is the hottest-spot conductor rise over top-liquid temperature under rated conditions (°C)
$P_{\mathrm{LL}}\left(\mathrm{pu}\right)$	is the per-unit load loss
$P_{\mathrm{LL-R}}\left(\mathrm{pu}\right)$	is the per-unit load loss under rated conditions

This may then be written as in Equation (21):

$$\theta_{\mathrm{g}} = \theta_{\mathrm{g-R}} \times \left(\frac{I^2(pu) \times \left(1 + F_{\mathrm{HL}} \times P_{\mathrm{EC\text{-}R}}\,(\mathrm{pu})\right)}{1 + P_{\mathrm{EC-R}}\,(\mathrm{pu})} \right)^{0.8} \tag{21}$$

where

θ_{g}	is the hottest-spot conductor rise over top-liquid temperature (°C)
$\theta_{\mathrm{g-R}}$	is the hottest-spot conductor rise over top-liquid temperature under rated conditions (°C)
F_{HL}	is the harmonic loss factor for winding eddy currents
$P_{\mathrm{EC-R}}\left(\mathrm{pu}\right)$	is the per-unit winding eddy-current loss under rated conditions

As an example, a 65 °C average winding rise, 80 °C hottest spot rise liquid immersed transformer was designed for a specified harmonic current content. After installation, the actual harmonic currents were measured and the current spectrum was supplied to the manufacturer with a request to check the temperature rises. At rated load and 60 Hz, the tested losses were as follows:

—No load: 4 072 W

—I^2R: 27 821 W

—Stray and eddy loss: 4 060 W

—Total loss: 35 953 W

The measured temperature rises above ambient were as follows:

—HV average rise: 48.1 °C

—LV average rise: 47.6 °C

—Top-liquid rise: 47.2 °C

—Hottest spot conductor rise: 55.3 °C

The harmonic distribution was determined at a load, which was approximately 100% of the magnitude of the fundamental current. The harmonic distribution, normalized to the fundamental, was supplied as in Table 10.

Table 10—Harmonic distribution for maximum load current Example 3

h	1	3	5	7	9	11	13	15	17	19	23	25
$\frac{I_h}{I_1}$	1.00	0.35	0.17	0.12	0.092	0.071	0.051	0.043	0.040	0.039	0.032	0.029

The calculations to determine the harmonic loss factors for the winding eddy losses and the other stray losses are shown in Table 11.

Table 11—Tabulated calculation of the harmonic loss factor for Example 3

h	$\frac{I_h}{I_1}$	$\left(\frac{I_h}{I_1}\right)^2$	h^2	$\left(\frac{I_h}{I}\right)^2 h^2$	$h^{0.8}$	$\left(\frac{I_h}{I_1}\right)^2 h^{0.8}$
1	1.00	1.0000	1	1.0000	1.0000	1.0000
3	0.35	0.12250	9	1.1025	2.4082	0.29501
5	0.17	0.028900	25	0.72250	3.6239	0.10473
7	0.12	0.014400	49	0.70560	4.7433	0.068303
9	0.092	0.084640	81	0.68558	5.7995	0.049087
11	0.071	0.0050410	121	0.60996	6.8095	0.034327
13	0.051	0.0026010	169	0.43957	7.7831	0.020244
15	0.043	0.0018490	225	0.41603	8.7272	0.016137
17	0.040	0.0016000	289	0.46240	9.6463	0.015434
19	0.039	0.0015210	361	0.54908	10.544	0.016037
23	0.032	0.0010240	529	0.54170	12.285	0.012580
25	0.029	0.00084100	625	0.52563	13.133	0.011045
Σ		1.189		7.761		1.643

The third column summation is 1.19. The square root of this number results in a per-unit rms current of 1.09. The fifth column summation divided by the third column summation results in a harmonic loss factor for winding eddy losses of 6.53. The seventh column summation divided by the third column summation results in a harmonic loss factor for other stray losses of 1.38.

An engineering analysis indicated the division of the eddy and other stray losses to be:

—Eddy loss: 316 W

—Other stray loss: 3744 W

—Total stray loss: 4060 W

In order to determine the top-liquid rise, the total losses must be corrected to reflect the higher rms current above the rated current and the effects of the harmonic content.

$$P_{\mathrm{LL}}(\mathrm{pu}) = P_{\mathrm{LL-R}}(\mathrm{pu}) \times (1.09)^2$$

where

$P_{\mathrm{LL}}(\mathrm{pu})$ is the per-unit load loss
$P_{\mathrm{LL-R}}(\mathrm{pu})$ is the per-unit load loss under rated conditions

The calculation of the total enhanced losses due to the harmonic load using Equation (19) is then tabulated as shown in Table 12.

Table 12—Tabulated loss calculation

Type of loss	Rated losses(W)	Load losses(W)	Harmonic multiplier	Corrected losses
No load	4072	4072		4072
I^2R	27 821	33 107		33 107
Winding eddy	316	376	6.53	2455
Other stray	3744	4455	1.38	6148
Total losses	35 953	42 010		45 782

The top-liquid rise for the specified loading conditions may now be calculated by Equation (18):

$$\theta_{\mathrm{TO}} = \theta_{\mathrm{TO-R}} \times \left(\frac{P_{\mathrm{LL}} + P_{\mathrm{NL}}}{P_{\mathrm{LL-R}} + P_{\mathrm{NL}}} \right)^{0.8} = 47.2 \times \left(\frac{45\ 782}{35\ 953} \right)^{0.8} = 57.3$$

where

θ_{TO} is the top-liquid-rise over ambient temperature (°C)
$\theta_{\mathrm{TO-R}}$ is the top-liquid-rise over ambient temperature under rated conditions (°C)
P_{LL} is the load loss (watts)
$P_{\mathrm{LL-R}}$ is the load loss under rated conditions (watts)
P_{NL} is the no load loss (watts)

The maximum per-unit eddy loss occurred in the high-voltage winding and was calculated to be an average of 2% of the ohmic loss. Assuming the maximum eddy loss at the hottest spot region to be four times[12] the average eddy loss would give an eddy loss of 8% of the ohmic loss density at the hottest spot location. The hottest spot conductor rise over top-liquid temperature can be calculated by Equation (21):

$$\theta_{\mathrm{g}} = \theta_{\mathrm{g-R}} \times \left(\frac{I^2(pu) \times (1 + F_{\mathrm{HL}} \times P_{\mathrm{EC\text{-}R}}(\mathrm{pu}))}{1 + P_{\mathrm{EC-R}}(\mathrm{pu})} \right)^{0.8} = (55.3 - 47.2) \times \left(\frac{1 + 6.53 \times 0.08}{1 + 0.08} \times 1.19 \right)^{0.8} = 12.3$$

where

θ_{g} is the hottest-spot conductor rise over top-liquid temperature (°C)
$\theta_{\mathrm{g-R}}$ is the hottest-spot conductor rise over top-liquid temperature under rated conditions (°C)
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{\mathrm{EC-R}}(\mathrm{pu})$ is the per-unit winding eddy-current loss under rated conditions

The hottest spot conductor rise over ambient then becomes:

$$57.3 + 12.3 = 69.6\ ^\circ\mathrm{C}$$

[12]See Annex F for an explanation for using four times the average eddy loss as an estimate of the maximum eddy loss density.

6.2 Transformer capability equivalent calculation using data available from certified test report

In order to make the calculation with limited data, certain assumptions shown in this clause are considered to be conservative. The estimates are also directed toward smaller transformers, generally less than 5000 kVA. Larger power transformers use various shielding techniques to control stray losses and the manufacturer should be consulted rather than using these general assumptions. Even for smaller transformers, these assumptions may be modified based on guidance from the manufacturer for a particular transformer.

a) The certified test report includes all data listed in the appendixes to IEEE Std C57.12.00 or IEEE Std C57.12.91.

b) A portion of the total stray loss, determined by the multipliers in Table 13 and Table 14 below, is assumed to be winding eddy-current loss. The remaining stray loss then is external to the windings and is noted in the tables as other stray loss. See Annex G for sample transformer test data for reference purposes.

c) The I^2R loss is assumed to be uniformly distributed in each winding.

d) The eddy-current loss distribution within each winding is assumed to be non-uniform.

e) The maximum eddy-current loss density is assumed to be in the region of the winding hottest spot and is assumed to be 400% of the average eddy-current loss density for that winding. Finite element analyses and empirical data indicate that smaller ratings may show a uniform distribution of eddy-current loss (Hwang [B26]).

Table 13—Estimate of distribution of total stray loss % for dry-type transformers

	Winding eddy loss		Other stray loss
	LV winding	HV winding	
Max self-cooled current rating < 1 000 A or ≤ 4:1 turns ratio	20	15	65
> 4:1 turns ratio	25	10	

Table 14—Estimate of distribution of total stray loss % for liquid immersed transformers

kVA range	Winding eddy loss		Other stray loss
	LV winding	HV winding	
≤ 300	55	5	40
> 300 ≤ 1 000	40	10	50
> 1 000 ≤ 3 000	20	10	70
> 3 000	25	15	60

CAUTION

These are conservative estimates in most cases and should not be followed if better data is available from the manufacturer.

As established in test codes IEEE Std C57.12.90 and IEEE Std C57.12.91, the stray loss component of the load loss is calculated by subtracting the I^2R loss of the transformer from the measured load loss.[13] Therefore, as in Equation (22):

$$P_{\text{TSL}-\text{R}} = P_{\text{LL}-\text{R}} - K \times \left[\left(I_{1-R} \right)^2 \times R_1 + \left(I_{2-R} \right)^2 \times R_2 \right] \tag{22}$$

where

$P_{\text{TSL}-\text{R}}$ is the total stray loss under rated conditions (watts)
$P_{\text{LL}-\text{R}}$ is the load loss under rated conditions (watts)
K is a constant dependent on the number of phases: 1.0 for single-phase transformers 1.5 for three-phase transformers
$I_{1-\text{R}}$ is the high-voltage (HV) rms fundamental line current under rated frequency and rated load conditions (amperes)
R_1 is the dc resistance measured between two HV terminals (ohms)
$I_{2-\text{R}}$ is the low voltage (LV) rms fundamental line current under rated frequency and rated load conditions (amperes)
R_2 is the dc resistance measured between two LV terminals (ohms)

By assumption b) of this sub clause, a portion of the stray loss is taken to be eddy-current loss. For example, for dry-type transformers, the winding-eddy loss is assumed to be as shown in Equation (23):

$$P_{\text{EC}-\text{R}} = P_{\text{TSL}-\text{R}} \times 0.35 \tag{23}$$

where

$P_{\text{EC}-\text{R}}$ is the winding eddy-current loss under rated conditions (watts)
$P_{\text{TSL}-\text{R}}$ is the total stray loss under rated conditions (watts)

For a 1000 kVA liquid immersed transformer, the winding eddy loss is assumed to be as shown in Equation (24):

$$P_{\text{EC}-\text{R}} = P_{\text{TSL}-\text{R}} \times 0.50 \tag{24}$$

where

$P_{\text{EC}-\text{R}}$ is the winding eddy-current loss under rated conditions (watts)
$P_{\text{TSL}-\text{R}}$ is the total stray loss under rated conditions (watts)

The other stray losses are then calculated as follows in Equation (25):

$$P_{\text{OSL}-\text{R}} = P_{\text{TSL}-\text{R}} - P_{\text{EC}-\text{R}} \tag{25}$$

where

$P_{\text{OSL}-\text{R}}$ is the other stray loss under rated conditions (watts)
$P_{\text{TSL}-\text{R}}$ is the total stray loss under rated conditions (watts)
$P_{\text{EC}-\text{R}}$ is the winding eddy-current loss under rated conditions (watts)

[13]Many test reports for three-phase transformers show the resistance of three phases in series. In these cases, values for R_1 and R_2 may be calculated as follows: a) Delta winding: R_1 or R_2 = 2/9 of three-phase resistance, and b) Wye winding: R_1 or R_2 = 2/3 of three-phase resistance.

The low-voltage (inner) winding eddy-current loss for a dry-type transformer can be calculated from the value of $P_{\text{EC-R}}$ determined from Equation (23) and Table 13, depending on the transformer turns ratio and current rating. The low-voltage (inner) winding eddy-current loss for a liquid immersed transformer can be calculated from the value of $P_{\text{EC-R}}$ determined from Equation (24) and Table 14, depending on the transformer kVA. Since by assumption c) in this clause, the I^2R loss is assumed to be uniformly distributed within the winding, and by assumption e), the maximum eddy-current loss density is assumed to be 400% of the average value, the low-voltage winding eddy-current loss in per-unit of that winding's I^2R loss for a low voltage general purpose dry-type transformer will be as shown in Equation (26):

$$\text{Max } P_{\text{EC}-\text{R}}(\text{pu}) = \frac{0.20 \times 4 \times P_{\text{EC}-\text{R}}}{K \times \left(I_{2-\text{R}}\right)^2 \times R_2} \tag{26}$$

For a 300 kVA liquid immersed transformer as shown in Equation (27):

$$\text{Max } P_{\text{EC}-\text{R}}(\text{pu}) = \frac{0.55 \times 4 \times P_{\text{EC}-\text{R}}}{K \times \left(I_{2-\text{R}}\right)^2 \times R_2} \tag{27}$$

where

$P_{\text{EC}-\text{R}}(\text{pu})$ is the per-unit winding eddy-current loss under rated conditions
$P_{\text{EC}-\text{R}}$ is the winding eddy-current loss under rated conditions (watts)
$I_{2-\text{R}}$ is the low voltage (LV) rms fundamental line current under rated frequency and rated load conditions (amperes)
R_2 is the dc resistance measured between two LV terminals (ohms)
K is a constant dependent on the number of phases: 1.0 for single-phase transformers, 1.5 for three-phase transformers

For the same example as above, the winding eddy losses for the outer or HV winding may be calculated in a similar manner. For the low voltage dry type general purpose transformer, the losses are calculated as shown in Equation (28):

$$\text{Max } P_{\text{EC}-\text{R}}(\text{pu}) = \frac{0.15 \times 4 \times P_{\text{EC}-\text{R}}}{K \times \left(I_{1-\text{R}}\right)^2 \times R_1} \tag{28}$$

For a 300 kVA liquid immersed transformer the losses are as shown in Equation (29):

$$\text{Max } P_{\text{EC}-\text{R}}(\text{pu}) = \frac{0.05 \times 4 \times P_{\text{EC}-\text{R}}}{K \times \left(I_{1-\text{R}}\right)^2 \times R_1} \tag{29}$$

where

$P_{\text{EC}-\text{R}}(\text{pu})$ is the per-unit winding eddy-current loss under rated conditions
$P_{\text{EC}-\text{R}}$ is the winding eddy-current loss under rated conditions (watts)
$I_{1-\text{R}}$ is the high voltage (HV) rms fundamental line current under rated frequency and rated load conditions (amperes)
R_1 is the dc resistance measured between two HV terminals (ohms)
K is a constant dependent on the number of phases: 1.0 for single-phase transformers, 1.5 for three-phase transformers

6.2.1 Typical calculations for dry-type transformers

A nonsinusoidal load current with harmonic distribution is shown in Table 15.

Table 15—Harmonic distribution for maximum load current Example 4

h	1	2	3	4	5	6	7	8	9	10	11	12	13
$\frac{I_h}{I_1}$	1.00	0.044	0.092	0.022	0.41	0.018	0.20	0.010	0.018	0.015	0.046	0.010	0.048

Determine the maximum load current that can be continuously drawn (under standard conditions) from an IEEE C57.12.01 dry-type transformer with the following characteristics taken from the certified test report:

- High-voltage winding
  - 13 800 V delta
  - Resistance $= 2.0679\ \Omega$ @ 100 °C*
- Low-voltage winding
  - 480 V wye
  - Resistance $= 0.000589\ \Omega$ @ 100 °C*
- Rated capacity
  - 2500 kVA
  - Three-phase
  - 80 °C rise
  - Type AA cooling class
- Load losses at 100 °C = 15 723 W
  - (*Resistances are the sum of the three phases in series.)
- Values for R_1 and R_2 can be determined using footnote 12 in 6.2:
  - $R_1 = 0.4595\ \Omega$
  - $R_2 = 0.000393\ \Omega$
- Values for $I_{1\text{-R}}$ and $I_{2\text{-R}}$ calculated from kVA and voltage ratings are as follows:
  - $I_{1\text{-R}} = 104.6$ A
  - $I_{2\text{-R}} = 3007$ A
- The total stray loss can be calculated from Equation (22) as follows:
  - $P_{\text{TSL-R}} = 15\ 723 - 1.5 \times (104.6P^{2\text{P}} \times 0.4595 + 3007P^{2\text{P}} \times 0.000393)$
  - $P_{\text{TSL-R}} = 15\ 723 - 1.5 \times (5027 + 3554)$
  - $P_{\text{TSL-R}} = 15\ 723 - 12\ 872 = 2851$ W
- The winding eddy loss is then calculated by assumption b) in 6.2 and by Equation (23):
  - $P_{\text{EC-R}} = 2\ 851 \times 0.35 = 998$ W

Since the transformer turns ratio exceeds 4:1 and the secondary current exceeds 1000 A, the low-voltage winding eddy-current loss is 0.25 times $P_{\text{EC–R}}$ and Max $P_{\text{EC–R}}$ can be calculated from Equation (26) as follows:

$$\text{Max } P_{\text{EC-R}}(\text{pu}) = \frac{1.0 \times 998}{1.5 \times 3554} = 0.187 \text{ pu}$$

As in the previous example, values for $I_h{}^2$(pu), h^2, and $I_h{}^2(\text{pu})_h{}^2$ are required for the calculation of P_{LL}(pu) from Equation (16). These values are calculated and tabulated as in Table 16.

Table 16—Tabulated calculation of the harmonic loss factor for Example 4

h	$\dfrac{I_h}{I_1}$	$\left(\dfrac{I_h}{I_1}\right)^2$	h^2	$\left(\dfrac{I_h}{I_1}\right)^2 h^2$
1	1.00	1.0000	1	1.0000
2	0.044	0.0019360	4	0.0077440
3	0.092	0.0084640	9	0.076176
4	0.022	0.00048400	16	0.0077440
5	0.41	0.16810	25	4.2025
6	0.018	0.00032400	36	0.011664
7	0.20	0.040000	49	1.9600
8	0.010	0.00010000	6	0.0064000
9	0.018	0.00032400	81	0.026244
10	0.015	0.00022500	100	0.022500
11	0.046	0.00211600	121	0.25604
12	0.010	0.00010000	144	0.014400
13	0.048	0.0023040	169	0.38938
Σ		1.224		7.981

According to Equation (8), the square root of the third column summation results in a per-unit rms value of 1.11 for the nonsinusoidal load current. The fifth column summation divided by the third column summation results in a harmonic loss factor of 6.52. From Equation (16), the local loss density produced by the nonsinusoidal load current in the region of highest eddy-current loss is as follows:

$$P_{\text{LL}}(\text{pu}) = I^2(\text{pu}) \times (1 + F_{\text{HL}} \times P_{\text{EC-R}}(\text{pu})) = 1.224 \times (1 + 6.52 \times 0.187) = 2.72 \text{ pu}$$

Thus, the rms value of the maximum permissible nonsinusoidal load current with the given harmonic composition, from Equation (17), is as follows:

$$I_{\max}(\text{pu}) = \sqrt{\frac{P_{\text{LL-R}}(\text{pu})}{1+F_{\text{HL}} \times P_{\text{EC-R}}(\text{pu})}} = \sqrt{\frac{1.187}{1 + 6.52 \times 0.187}} = 0.731$$

where

$I_{\max}(\text{pu})$ is the maximum permissible rms nonsinusoidal load current under rated conditions
$P_{\text{LL-R}}(\text{pu})$ is the per-unit load loss under rated conditions
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{\text{EC-R}}(\text{pu})$ is the per-unit winding eddy-current loss under rated conditions

In this case, the transformer capability with the given nonsinusoidal load current harmonic composition is approximately 70.5% of its sinusoidal load current capability.

$$I_{\max} = 0.731 \times 3007 = 2198 \text{ A}$$

6.2.2 Typical calculations for liquid immersed transformers

The next example illustrates the corrected temperature rise calculations for a liquid immersed transformer, meeting IEEE Std C57.12.00, with the following characteristics taken from the certified test report:

- High-voltage winding
  - 34 500 V delta
  - Resistance $= 18.207\ \Omega\ @\ 75\ ^{\circ}\mathrm{C}*$
- Low-voltage winding
  - 2400 V wye
  - Resistance $= 0.02491\ \Omega\ @\ 75\ ^{\circ}\mathrm{C}*$
- Rated capacity
  - 2500 kVA
  - Three-phase
  - 55 °C average winding rise
  - 65 °C hottest-spot rise
- Type ONAN cooling class
- No-load losses = 5100 W
- Load losses at 75 °C = 21 941 W
  - (*Resistances are the sum of the three phases in series.)
- Values for R_1 and R_2 can be determined using footnote 12 in 6.2:
  - $R_1 = 4.046\ \Omega$
  - $R_2 = 0.01661\ \Omega$
- Values for $I_{1\text{-R}}$ and $I_{2\text{-R}}$ calculated from kVA and voltage ratings are as follows:
  - $I_{1\text{-R}} = 41.8$ A
  - $I_{2\text{-R}} = 601.4$ A
- The total stray loss can be calculated from Equation (22) as follows:
  - $P_{\mathrm{TSL-R}} = 21\,941 - 1.5 \times \left(41.8^2 \times 4.046 + 601.4^2 \times 0.01661\right)\ \mathrm{W}$
  - $P_{\mathrm{TSL-R}} = 21\,941 - 1.5 \times (7069 + 6008)\ \mathrm{W}$
  - $P_{\mathrm{TSL-R}} = 21\,941 - 19\,615 = 2326\ \mathrm{W}$
- The winding eddy loss is then calculated by the assumption in Table 14 in 6.2.
  - $P_{\mathrm{EC-R}} = 2326 \times 0.30 = 698\ \mathrm{W}$
- By Equation (25), the other stray losses are as follows:
  - $P_{\mathrm{OSL-R}} = 2326 - 698 = 1628\ \mathrm{W}$

The data may be tabulated as follows:

- No load: 5100 W
- I^2R: 19 615 W
- Stray and eddy loss: 2326 W

—Total loss: 27 041 W

The assumed temperature rises above ambient are as follows:

—HV and LV average rise: 55 °C

—Top-liquid rise: 55 °C

—Hottest spot conductor rise: 65 °C

The harmonic distribution was determined at a load that was approximately 75% of the magnitude of the fundamental current. The distribution, normalized to the fundamental, was supplied as in Table 17.

Table 17—Harmonic distribution for maximum load current Example 5

h	1	3	5	7	9	11	13	15	17	19
$\frac{I_h}{I_1}$	1.00	0.45	0.27	0.19	0.092	0.071	0.051	0.043	0.040	0.039

The calculations to determine the harmonic loss factors for the winding eddy losses and the other stray losses are tabulated as in Table 18.

Table 18—Tabulated calculation of the harmonic loss factor for Example 5

h	$\frac{I_h}{I_1}$	$\left(\frac{I_h}{I_1}\right)^2$	h^2	$\left(\frac{I_h}{I_1}\right)^2 h^2$	$h^{0.8}$	$\left(\frac{I_h}{I_1}\right)^2 h^{0.8}$
1	1.00	1.00	1	1.00	1.0000	1.0000
3	0.45	0.2025	9	1.82	2.4082	0.48767
5	0.27	0.0729	25	1.82	3.6239	0.26418
7	0.19	0.0361	49	1.77	4.7433	0.17123
9	0.092	0.0085	81	069	5.7995	0.049087
11	0.071	0.0050	121	0.61	6.8095	0.034327
13	0.051	0.0026	169	0.44	7.7831	0.020244
15	0.043	0.0018	225	0.42	8.7272	0.016137
17	0.040	0.0016	289	0.46	9.6463	0.015434
19	0.039	0.0015	361	0.55	10.544	0.016037
Σ		1.333		9.577		2.074

The third column summation is 1.33. The square root of this number results in a per-unit rms current of 1.15. The fifth column summation divided by the third column summation results in a harmonic loss factor for winding eddy losses of 7.18. The seventh column summation divided by the third column summation results in a harmonic loss factor for other stray losses of 1.55.

The division of the eddy and other stray losses is tabulated as follows:

—Eddy loss: 698 W

—Other stray loss: 1628 W

—Total stray loss: 2326 W

In order to determine the top-liquid rise, the total losses must be corrected to reflect the lower rms current below the rated current and also the effects of the harmonic content. The rms current corrected for the 75% load results in the following multiplier to determine losses at the specified load conditions:

$$P_{\mathrm{LL}}(\mathrm{pu}) = 1.15^2 \times 0.75^2 = 0.74$$

Equation (19) is tabulated as in Table 19.

Table 19—Tabulated loss calculation

Type of loss	Rated losses (W)	Load losses (W)	Harmonic multiplier	Corrected losses
No load	5100	5100		5100
I^2R	19 615	14 515		14 515
Winding eddy	698	516	7.184	3707
Other stray	1628	1205	1.556	1875
Total losses	27 041	21 336		25 197

The top-liquid rise may now be calculated by Equation (18), as follows:

$$\theta_{\mathrm{TO}} = 55 \times \left(\frac{25761}{27041}\right)^{0.8} = 52.0\,°\mathrm{C}$$

The rated inner or LV winding losses can be calculated as follows:

$$I_{2-\mathrm{R}}{}^2 R = 1.5 \times 601.4^2 \times 0.01661 = 9011\ \mathrm{W}$$

The losses under the specified load conditions are:

$$I_2{}^2 R = 9011 \times (1.15 \times 0.75)^2 = 6703\ \mathrm{W}$$

By Table 14 in 6.2, it is assumed that 20% of the winding eddy losses are in the LV winding. The maximum eddy loss at the hottest spot region is assumed to be four times the average eddy loss. The hottest spot conductor rise over top-liquid temperature can be calculated by Equation (21), using watts rather than per-unit values.

$$\theta_{\mathrm{g}} = (65 - 55) \times \left(\frac{6703 + 3707 \times 0.2 \times 4}{19\,615 + 698}\right)^{0.8} = 10 \times \left(\frac{6703 + 2966}{20\,313}\right)^{0.8} = 5.52\,°\mathrm{C}$$

The hottest spot conductor rise over ambient then becomes:

$$52.0 + 5.52 = 57.5\ °\mathrm{C}$$

6.3 Neutral bus capability for nonsinusoidal load currents that include third harmonic components

The presence of third harmonic components in the nonsinusoidal load current composition can introduce zero-sequence currents into the neutral bus of a wye-connected transformer. The summing of third harmonic zero sequence currents in the neutral bus of the wye connected winding of the transformer causes increased heating that can exceed the thermal rating of the transformer neutral bus. When third harmonic currents are found to be present in the load current of a wye connected transformer, the capacity of the neutral bus should be calculated (see Hiranandani [B24]) based on the triplen harmonic content of the load.

When third harmonic components are found to be present in the nonsinusoidal load current for wye connected transformers, a measurement of the neutral bus ampacity is recommended to determine the magnitude of the zero-sequence current. The transformer manufacturer may then be consulted to determine the capability of the neutral bus to carry the zero-sequence current.

Annex A

(informative)

Bibliography

Bibliographical references are resources that provide additional or helpful material but do not need to be understood or used to implement this standard. Reference to these resources is made for informational use only.

[B1] Arslan, E., M. E. Balci, and M. H. Hocaoglu, "An analysis into the effect of voltage harmonics on the maximum loading capability of transformers," IEEE 16th International Conference on Harmonics and Quality of Power (ICHQP), pp. 616-620, 2014.

[B2] Balci, M. E. and S. Sakar, "Optimal Design of Single-Tuned Passive Filters to Minimize Harmonic Loss Factor," Middle East Journal of Scientific Research, vol. 21, no. 11, pp. 2149-2155, 2014.

[B3] Bewley, L. V., Two Dimensional Fields in Electrical Engineering. New York: McMillan, 1948.

[B4] Bishop, M. T., J. F. Baranowski, D. Heath, and S. J. Benna, "Evaluating harmonic-induced transformer heating," IEEE Transactions on Power Delivery, vol. 11, no. 1, pp. 305-311, 1996, http://dx.doi.org/10.1109/61.484029.

[B5] Bishop, M. T. and C. Gilker, "Harmonic caused transformer heating evaluated by a portable PC-controlled meter," 37th Annual Rural Electric Power Conference, 1993.

[B6] Blume, L. F. et al., Transformer Engineering, 2nd Ed. New York: Wiley, 1951, pp. 56-65.

[B7] Cox, M. D. and A. W. Galli, "Temperature rise of small oil-filled distribution transformers supplying nonsinusoidal load currents," IEEE Transactions on Power Delivery, vol. 11, no. 1, pp. 283-291, January 1996, http://dx.doi.org/10.1109/61.484027.

[B8] Crepaz, S., "Eddy current losses in rectifier transformers," IEEE Transactions on Power Apparatus and Systems, vol. PAS-89, no. 7, pp. 1651-1656, September/October 1970, http://dx.doi.org/10.1109/TPAS.1970.292813.

[B9] Dalila, M. S., M. N. Khalid, and M. Md Shah, "Distribution transformer losses evaluation under non-linear load," Australasian Universities Power Engineering Conference, 2009, pp. 1-6.

[B10] Driesen, J., T. Van Craenenbroeck, B. Brouwers, K. Hameyer, and R. Belmans, "Practical method to determine additional load losses due to harmonic currents in transformers with wire and foil windings," Proceedings of the IEEE Winter Meeting, Singapore, Jan. 2000.

[B11] Dwight, H. B., Electrical Coils and Conductors. New York: The McGraw-Hill Companies, 1945, Ch. 3.

[B12] Elmoudi, A., M. Lehtonen, and H. Nordman, "Corrected winding eddy-current harmonic loss factor for transformers subject to nonsinusoidal load currents," in Power Tech, IEEE Russia. IEEE, 2005, pp. 1-6.

[B13] Elmoudi, A. and M. Lehtonen, "Eddy losses calculation in transformer windings using FEM," The 44th International Scientific Conference of Riga Technical University, Riga, Latvia, pp. 46-51, 2003.

[B14] Elmoudi, A., M, Lehtonen, and H. Nordman, "Effect of harmonics on transformers loss of life," Conference Record of the 2006 IEEE International Symposium on Electrical Insulation, pp. 408-411, 2006, http://dx.doi.org/10.1109/ELINSL.2006.1665344.

[B15] Ferguson, G. N. C., "Power quality improvement in a harmonic environment," *International Testing Association (NETA)* Annual Technical Conference, St. Louis, MO, Mar. 19, 1997.

[B16] Field, A. B., "Eddy currents in large slot-wound conductors," *AIEE Proceedings*, vol. 24, 1905.

[B17] Frank, J. M., "Origin, development and design of K factor transformers," IEEE Industry Applications Magazine, vol. 3, no. 5, pp. 67-69, September/October 1997, http://dx.doi.org/10.1109/2943.612239.

[B18] Fuchs, E. F., D. Yildirim, and W. Mack Grady, "Measurement of eddy-current loss coefficient P_{EC-R}, derating of single-phase transformers, and comparison with K-factor approach," IEEE Transactions on Power Delivery, vol. 15, no. 1, pp. 148-154, 2000, http://dx.doi.org/10.1109/61.847243.

[B19] Gholamian, S. A., "Life Estimation of Distribution Transformers Under Non-Linear Loads Using Calculated Loss by 2D-FEM," J. Electrical Systems, vol. 7, no. 1, pp. 12-24, 2011.

[B20] Gómez, J. C. and M. M. Morcos, "Impact of EV battery chargers on the power quality of distribution systems," IEEE Transactions on Power Delivery, vol. 18, no. 3, pp. 975-981, 2003, http://dx.doi.org/10.1109/TPWRD.2003.813873.

[B21] Gumilang, H. and A. Suslilo, "Effect of power system harmonic on degradation process of transformer insulation system," *IEEE 9th International Conference on the Properties and Applications of Dielectric Materials*, pp. 261-264.

[B22] Gupta, A. and R. Singh, "Computation of transformer losses under the effects of non-sinusoidal currents," Advanced Computing: An International Journal, vol. 2, no. 6, pp. 91-104, 2011.

[B23] Gupta, A. and R. Singh, "Evaluation of Distribution Transformer Losses Under Harmonic Loads Using Analytical and Simulation Methods," International Journal on Emerging Technologies, vol. 2, no. 2, pp. 90-95, 2011.

[B24] Hesterman, B., "Time-domain K-factor computation methods," 29th International Power Conversion Conference, Sept. 1994.

[B25] Hiranandani, A., "Calculation of ampacities and sizing of line and neutral conductors in the presence of harmonics," Transactions of IEEE Insulated Conductors Committee, St. Petersburg, FL, Fall 2000.

[B26] Hwang, M. D., W. M. Grady, and H. W. Sanders Jr., "Assessment of winding losses in transformers due to harmonic currents," International Conference on Harmonics in Power Systems, Worcester Polytechnic Institute, Worcester, MA, Oct. 1984.

[B27] Hwang, M. D., W. M. Grady, and H. W. Sanders Jr., "Calculation of winding temperatures in distribution transformers subjected to harmonic currents," IEEE Transactions on Power Delivery, vol. 3, no. 3, pp. 1074-1079, July 1988, http://dx.doi.org/10.1109/61.193889.

[B28] Hwang, M. D., W. M. Grady, and H. W. Sanders Jr., "Distribution transformer winding losses due to nonsinusoidal currents," IEEE Transactions on Power Delivery, vol. PWRD-2, no. 1, pp. 140-146, January 1987, http://dx.doi.org/10.1109/TPWRD.1987.4308084.

[B29] IEC 61378-2006, Converter Transformers—Part 1: Transformers for Industrial Applications.[14]

[B30] IEEE Std C57.12.80™-1978, IEEE Standard Terminology for Power and Distribution Transformers.[15,16]

[B31] IEEE Std C57.18.10™-1998, IEEE Standard Practices and Requirements for Semiconductor Power Rectifier Transformers.

[B32] Jayasinghe, R. N., J. R. Lucas, and K. B. I. M. Perera, "Power system harmonic effects on distribution transformers and new design considerations for K factor transformers," IEE Sri Lanka Annual Sessions, 2003.

[B33] Kaul, H. J., "Stray-current losses in stranded windings of transformers," *Power Apparatus and Systems, Part III*," Transactions of the American Institute of Electrical Engineers, vol. 76, no. 3, pp. 137-146, 1957.

[B34] Kelley, A. W., S. W. Edwards, J. P. Rhode, and M. Baran, "Transformer derating for harmonic currents: A wideband measurement approach for energized transformers," IEEE Industry Applications Conference, vol. 1, pp. 840-847, Oct. 1995.

[B35] Kennedy, S. P., "Design and application of semiconductor rectifier transformers," IEEE Transactions on Industry Applications, vol. 38, no. 4, pp. 927-933, July/August 2002, http://dx.doi.org/10.1109/TIA.2002.800595.

[B36] Kennedy, S. P. and C. L. Ivey, "Application, design and rating of transformers containing harmonic currents," 1990 Annual Pulp and Paper Industry Technical Conference, http://dx.doi.org/10.1109/PAPCON.1990.109859.

[B37] Kennedy, S. P. and T. Gordner, "Hot spot studies for sheet wound transformer windings," 80th International Conference of Doble Clients, 2013.

[B38] Key, T. and J. Lai, "Costs and benefits of harmonic current reduction for switch-mode power supplies in a commercial office building," *IEEE IAS Annual Meeting*, Orlando, FL, Oct. 1995.

[B39] Laminar, J. and M. Stafl, Eddy Currents. London: Iliffe, 1966, p. 33.

[B40] Makarov, S. N. and A. E. Emanuel, "Corrected harmonic loss factor for transformers supplying nonsinusoidal load currents," Proceedings of the Ninth International Conference on Harmonics and Quality of Power, pp. 87-90, 2000, http://dx.doi.org/10.1109/ICHQP.2000.897003.

[B41] Masoum, M. A. S., P. S. Moses, and A. S. Masoum, "Derating of asymmetric three-phase transformers serving unbalanced nonlinear loads," IEEE Transactions on Power Delivery, vol. 23, no. 4, pp. 2033-2041, 2008, http://dx.doi.org/10.1109/TPWRD.2008.923057.

[B42] Massey, G. W., "Estimation methods for power system harmonic effects on power distribution transformers," 37th Annual Rural Electric Power Conference, 1993.

[B43] Electrical Engineering Staff, M. I. T., Magnetic Circuits and Transformers. New York: Wiley, 1949, Ch. 5.

[14]IEC publications are available from the Sales Department of the International Electrotechnical Commission, Case Postale 131, 3, rue de Varembé, CH-1211, Genève 20, Switzerland/Suisse (http://www.iec.ch/). IEC publications are also available in the United States from the Sales Department, American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036, USA (http://www.ansi.org/).States from the Sales Department, American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036, USA (http://www.ansi.org/).
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[16]The IEEE standards or products referred to in this clause are trademarks of The Institute of Electrical and Electronics Engineers, Inc.

[B44] Parentoni, M. F. C., J. P. G. Abreu, and Fernando N. Belchior, "Design proposition of transformers used for feeding nonlinear loads—A practical approach based on temperature measurement," 14th International Conference on Harmonics and Quality of Power, pp. 1-8, 2010.

[B45] Pierce, L. W., "Transformer design and application considerations for nonsinusoidal load currents," IEEE Transactions on Industry Applications, vol. 31, no. 3, pp. 633-645, May/June 1996, http://dx.doi.org/10.1109/28.502176.

[B46] Sadati, S. B., A. Tahani, B. Darvishi, M. Dargahi, and H. Yousefi, "Comparison of distribution transformer losses and capacity under linear and harmonic loads," IEEE Power and Energy Conference, pp. 1265-1269, 2008.

[B47] Sadati, S. B., A. Tahani, M. Jafari, and M. Dargahi, "11th International Conference on Derating of Transformers under Non-sinusoidal Loads in Optimization of Electrical and Electronic Equipment," pp. 263-268, IEEE, 2008.

[B48] Said, D. M. and K. M. Nor, "Effects of harmonics on distribution transformers," Australasian Universities Power Engineering Conference, pp. 1-5, 2008.

[B49] Savaghebi, M., A. Gholami, and A. Jalilian, "Transformer dynamic loading capability assessment under nonlinear load currents," 43rd International Universities Power Engineering Conference, pp. 1-5, 2008.

[B50] Sharifian, M. B. B., J. Faiz, S. A. Fakheri, and A. Zraatparvar, "Derating of distribution transformers for non-sinusoidal load currents using finite element method," Proceedings of the 10th IEEE International Conference on Electronics, Circuits and Systems, 2003, vol. 2, pp. 754-757.

[B51] Taheri, S., H. Taheri, I. Fofana, H. Hemmatjou, and A. Gholami, "Effect of power system harmonics on transformer loading capability and hot spot temperature," 25th Canadian Conference on Electrical & Computer Engineering (CCECE), pp. 1-4, 2012.

[B52] UL 1561-2011, Dry-Type General Purpose and Power Transformers.[17]

[B53] UL 1562-2013, Transformers, Distribution, Dry-Type-Over 600 Volts.

[B54] Yildirim, D. and E. F. Fuchs, "Measured transformer derating and comparison with harmonic loss factor (F_{HL}) approach," IEEE Transactions on Power Delivery, vol. 15, no. 1, pp. 186-191, 2000, http://dx.doi.org/10.1109/61.847249.

[17]UL standards are available from Global Engineering Documents, 15 Inverness Way East, Englewood, CO 80112, USA (http://global.ihs.com/).

Annex B

(informative)

Tutorial discussion of transformer losses and the effect of harmonic currents on these losses

Power transformers with ratings up to 50 MVA are almost always of core form construction. High-voltage and low-voltage windings are concentric cylinders surrounding a vertical core leg of a rectangular or circular cross section. The vertical core legs and the horizontal core yoke members that constitute the magnetic circuit are made up of thin steel laminations. In the top and bottom yoke regions, there are usually external clamping structures (clamps) that may be made of either metallic or insulating materials. Liquid-immersed transformers are contained within a steel tank, whereas dry-type transformers may be either freestanding or surrounded by a metal enclosure. If direct current is passed through the transformer winding conductors, a simple I^2R loss will be produced, where R is the dc resistance of the winding. However, if an alternating current (ac) of the same magnitude is passed through the winding conductors, an additional loss is produced. This can be explained as follows. When the transformer windings carry the ac current, each conductor is surrounded by an alternating electromagnetic field whose strength is directly proportional to the magnitude of the current. A picture of the composite field produced by rated load current flowing through all the winding conductors is shown in Figure B.1, which is a cross-sectional view through the core, windings, clamps, and tank. Each metallic conductor linked by the electromagnetic flux experiences an internal induced voltage that causes eddy currents to flow in that conductor. The eddy currents produce losses that are dissipated in the form of heat, producing an additional temperature rise in the conductor over its surroundings. This type of extra loss beyond the I^2R loss is frequently referred to as stray loss. Although all of the extra loss is an eddy-current loss, the portion in the windings is usually called eddy-current loss (P_{EC}), and the portion outside the windings is called “other stray loss” (P_{OSL}).

Eddy-current loss in winding conductors is proportional to the square of the electromagnetic field strength (or the square of the load current that produces the field) and to the square of the ac frequency. Other stray losses are generally proportional to current raised to a power slightly less than 1, because the depth of penetration of the electromagnetic flux into the other metallic parts (usually steel) varies with the field strength. (For very high-frequency harmonic currents, the electromagnetic flux may not totally penetrate the winding conductors either, but it is conservative to assume that the eddy-current loss P_{EC} is proportional to the square of the harmonic current frequency.) When a transformer is subjected to a load current having significant harmonic content, the extra eddy-current loss in winding conductors and in other metallic parts will elevate the temperature of those parts above their normal operating temperature under rated conditions. Experience has shown that the winding conductors are the more critical parts for determination of acceptable operating temperature, so the objective should be to prevent the losses in winding conductors under harmonic load conditions from exceeding the losses under rated frequency operating conditions.

The inner winding of a core form transformer typically has higher eddy-current loss than the outer winding, because the electromagnetic flux has a greater tendency to fringe inwardly toward the low reluctance path of the core leg. Furthermore, the highest local eddy-current loss usually occurs in the end conductors of the inside winding. This is a result of the fact that this is the region of highest radial electromagnetic flux density (closest spacing of the radially directed flux lines as shown in Figure B.2), and the radial flux passes through the width dimension of the rectangular winding conductor. Since the width dimension of a conductor is typically three to five times the thickness dimension and eddy-current loss is proportional to the square of the dimension, high loss is produced in the end conductors. Certain simplifying assumptions have been made in this recommended practice about the relative proportions of the eddy-current losses in the inner and outer windings and the relation between average eddy-current losses and maximum local eddy-current losses. These assumptions, which are conservative, may be used when specific knowledge of the eddy-current loss magnitude is not available. However, more accurate calculations can be made if design values of eddy-current losses are available from the transformer manufacturer.

The recommendations for determination of acceptable operating conditions contained in this recommended practice are based on the calculation of a transformer capability equivalent, which establishes a current derating factor for load currents having a given harmonic composition. Equation) provides a calculation of the maximum rms value of a nonsinusoidal load current (in per unit of rated load current) that will help ensure that the losses in the highest loss density region of the windings do not exceed the design value of losses under rated frequency operating conditions. Example cases are presented for the situations where design eddy-current loss data are available from the manufacturer or where they are not.

Harmonic currents flowing through transformer leakage impedance and through system impedance may also produce some small harmonic distortion in the voltage waveform at the transformer terminals. Such voltage harmonics also cause extra harmonic losses in the transformer core. However, operating experience has not indicated that core temperature rise will ever be the limiting parameter for determination of safe magnitudes of nonsinusoidal load currents.

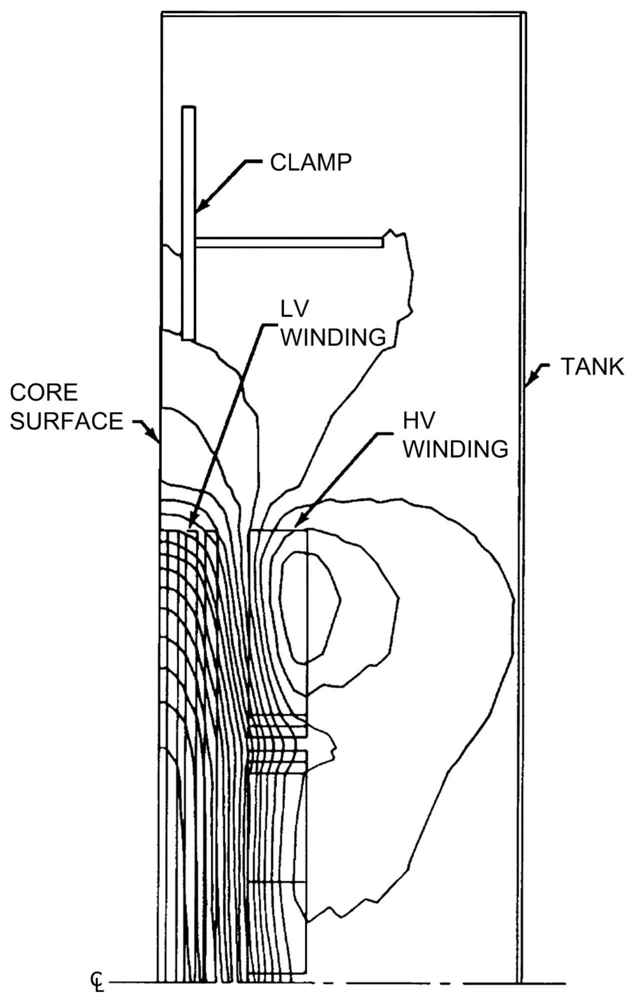

Figure B.1—Electromagnetic field produced by load current in a transformer

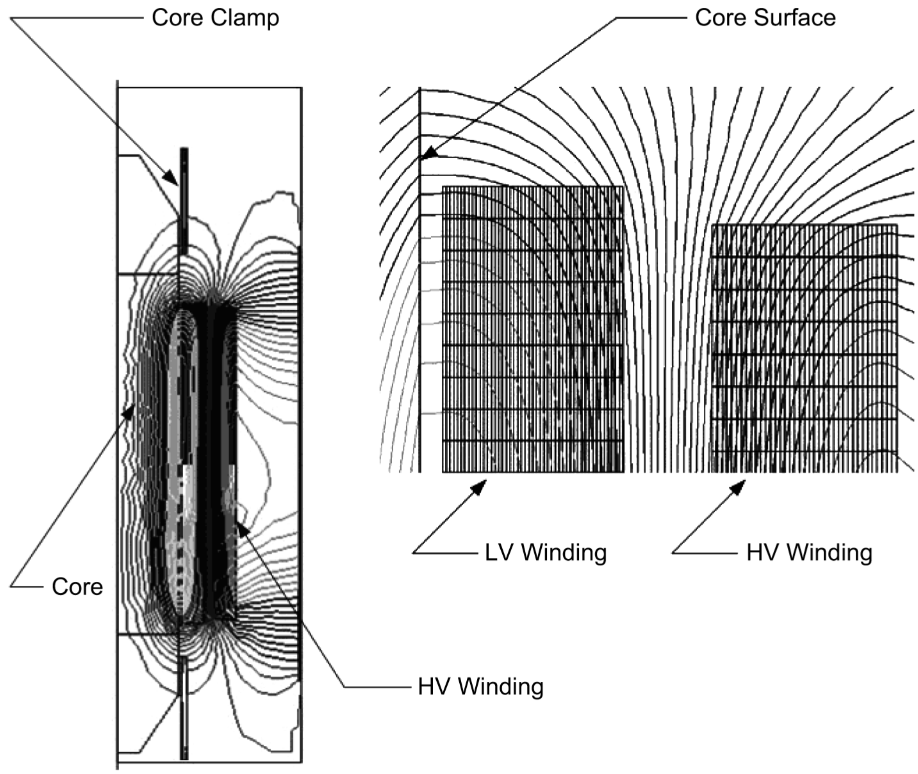

Figure B.2—Electromagnetic field in windings produced by load current in a transformer

Annex C

(informative)

Corrected harmonic loss factor for high frequencies[18]

C.1 Transformer winding losses computation using FEM

Two-dimensional finite element method is the most commonly used method to compute eddy losses. The analysis of the winding eddy loss by 3-D FEM will be more accurate, however the computation cost and the complexity increase and in many cases a reduction to a 2-D is possible without significant loss of accuracy.

For windings in which the conductors are small strands i.e., smaller than the depth of penetration, we can model the region as non-conducting region; find the resistance limited eddy current losses. The field solution gives the familiar leakage flux where the flux in the windings flows axially up through the coils and then bends radially across the windings. The leakage flux has its greatest concentration on the interface between the two windings and decreases, as progress is away from the gap between the windings. The inner LV winding typically has a higher attraction of the leakage flux due to the high permeance of the core. The HV winding divides its leakage flux with part being attached to the core and the reminder attracted to the core clamps and other structural parts.

The critical point in transformers is usually in the upper end of the windings where the conductors are exposed to an inclined magnetic field with two components, an axial component and a radial component, and the eddy current losses are the contribution of these two components. Using local flux density from FEM solution the eddy loss density can be calculated. In order to calculate the winding losses the P_{dc} loss must be added to the eddy loss [B13].

C.2 Corrected harmonic loss factor

At high frequencies conductor skin effect or the limited magnetic flux penetration has to be taken into account. A corrected harmonic loss factor is suggested based on the following Equation (C.1).

$$P_{\mathrm{EC}} = \frac{\pi^2 f^2 T^2 B^2}{3\rho} F(\xi) \tag{C.1}$$

where

$F(\xi)$ is equal to $\dfrac{3}{\xi}\dfrac{\sinh\xi - \sin\xi}{\cosh\xi - \cos\xi}$

$\xi = \dfrac{T}{\delta}$ is strand dimension related to skin depth

$\delta_{\mathrm{R}} = \sqrt{\dfrac{\rho}{\mu\pi f}}$ is the depth of penetration at power frequency 50 Hz or 60 Hz

$\delta = \dfrac{\delta_{\mathrm{R}}}{\sqrt{h}}$ is the depth of penetration at harmonic frequency

The function $F(\xi)$ is as shown in Figure C.1. For small values of ξ, $F(\xi) \approx 1$ and the loss at low frequencies is proportional to square of the frequency and the square of the conductor thickness.

[18] This annex is a summary of the paper by Elmoudi, Lehtonen, and Nordman [B12].

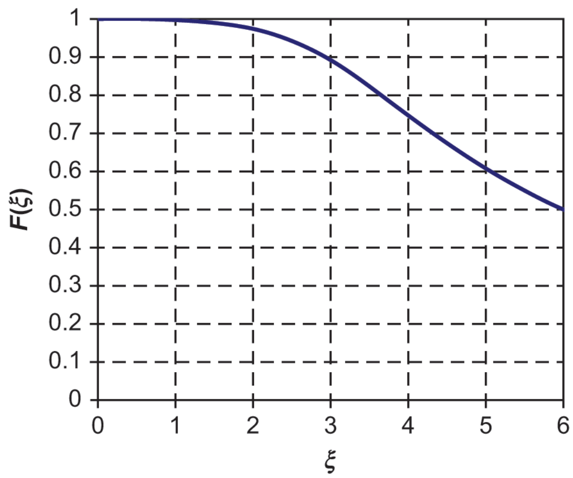

Figure C.1—The function *F*(*ξ*)

For copper conductor at 60 Hz and 75 °C $\delta_R \approx 9.4$ mm and for aluminum $\delta_R \approx 12.1$ mm. The strand dimension related to skin depth for a harmonic order *h* is:

$$\xi h = \frac{T}{\frac{\delta_R}{\sqrt{h}}} = \xi_R \sqrt{h} \tag{C.2}$$

as a function of the current

$$P_{EC} = KF(\xi)h^2 I^2 \tag{C.3}$$

when the current is non-sinusoidal

$$P_{EC} = K\sum_{h=1}^{h=h_{max}} F\left(\xi\right)h^2 I_h^2 \tag{C.4}$$

Normalizing the eddy current loss produced by a nonsinusoidal load current, to the eddy current losses at rated condition, results in a corrected harmonic loss factor.

$$F_{HL} = \frac{\sum_{h=1}^{h=h_{max}} \frac{F\left(\xi_h\right)}{F\left(\xi_R\right)} h^2 I_h^2}{\sum_{h=1}^{h=h_{max}} I_h^2} = \frac{\sum_{h=1}^{h=h_{max}} \frac{F\left(\xi_h\right)}{F\left(\xi_R\right)} h^2 \frac{I_h^2}{I^2}}{\sum_{h=1}^{h=h_{max}} \frac{I_h^2}{I^2}} \tag{C.5}$$

The importance of the above corrected factor is demonstrated in Figure C.2 where the graphs are presented as a function of the harmonic order, for different copper conductor thicknesses. The graph of the square harmonic order h^2 is added for comparison. For small conductors or at low harmonics the corrected harmonic loss factor approaches h^2.

Figure C.2—The function $h^2F(\xi_h)/F(\xi_R)$ versus harmonic order *h* for different rectangular copper conductor thickness

It can be seen that for small conductors skin effect is insignificant and the assumed h^2 harmonic loss factor predicts the losses accurately. The assumed loss factor predicts losses which are higher than the actual case for large conductors and higher harmonics due to the ignored conductor penetration depth. The actual flux impinging on the conductor will be less due to skin effect. The dimension of conductor sides used in transformers may be found in a wide range. The corrected harmonic loss factor presented can be used to count for both axial and radial eddy current.

C.3 Example calculation for dry-type transformer

The example in 6.1.1 is recalculated to compare the presented corrected harmonic loss factor with the harmonic order square and the harmonic loss factor suggested in this document: it is a standard transformer with 1200 A rated current and $P_{\text{EC-R}} = 0.15$ pu. The secondary winding is assumed to consist of copper strands of 3.65 mm × 11 mm, given the non-sinusoidal load current with the harmonic distortion as shown in Table C.1.

C.3.1 Harmonic order squared

The normalized current spectrum is shown in Table C.1.

Table C.1—Normalized current spectrum and calculated data for harmonic loss factor F_{HL}

h	$\frac{I_h}{I}$	$\left(\frac{I_h}{I}\right)^2$	h^2	$\left(\frac{I_h}{I}\right)^2 h^2$
1	1.00	1.0000	1	1.0000
5	0.233	0.054289	25	1.3572
7	0.11	0.011664	49	0.57154
11	0.042	0.0017640	121	0.21344
13	0.027	0.00072900	169	0.12320

Table continues

Table C.1—Normalized current spectrum and calculated data for harmonic loss factor F_{HL} *(continued)*

h	$\frac{I_h}{I}$	$\left(\frac{I_h}{I}\right)^2$	h^2	$\left(\frac{I_h}{I}\right)^2 h^2$
17	0.013	0.00016900	289	0.048841
19	0.0080	0.000064000	361	0.023104
Σ		1.0687		3.337

The harmonic loss factor F_{HL} is:

$$F_{\mathrm{HL}} = \frac{3.337}{1.0687} = 3.123 \text{ pu}$$

The total specific normalized losses are:

$$P_W\left(\text{pu}\right) = 1.0687 \times \left(1 + 0.15 \times 3.123\right) = 1.569 \text{ pu}$$

The value of the maximum non-sinusoidal current yielding the same winding eddy losses as sinusoidal current is:

$$I_{\max}\left(\text{pu}\right) = \sqrt{\frac{1.15}{1 + 0.15 \times 3.123}} = 0.885 \text{ pu}$$

or

$$I_{\max} = 0.885 \times 1200 = 1062\ A \text{ W}$$

C.3.2 Corrected harmonic loss factor

The normalized current spectrum using the corrected harmonic loss factor is shown in Table C.2.

Table C.2—Normalized current spectrum and the calculated data for the corrected harmonic loss factor F_{HL}

h	$\frac{I_h}{I}$	$\left(\frac{I_h}{I}\right)^2$	$\frac{h^2 F\left(\xi_h\right)}{F\left(\xi_R\right)}$	$\left(\frac{h^2 F\left(\xi_h\right)}{\xi_R}\right)\left(\frac{I_h}{I}\right)^2$
1	1.00	1.0000	1	1.0000
5	0.233	0.054289	23.39	1.2698
7	0.11	0.011664	43.27	0.50470
11	0.042	0.0017640	93.29	0.16456
13	0.027	0.00072900	121.59	0.08864
17	0.013	0.00016900	182.89	0.030908
19	0.0080	0.000064000	215.70	0.013805
Σ		1.0687		3.072

The harmonic loss factor F_{HL} is:

$$F_{\mathrm{HL}} = \frac{3.072}{1.0687} = 2.875 \text{ pu}$$

The total specific normalized losses are:

$$P_W\left(pu\right) = 1.0687 \times \left(1 + 0.15 \times 2.875\right) = 1.529 \text{ pu}$$

The value of the maximum non-sinusoidal current yielding the same winding eddy losses as sinusoidal current is:

$$I_{\max}\left(pu\right) = \sqrt{\frac{1.15}{1 + 0.15 \times 2.875}} = 0.896 \text{ pu}$$

or

$$I_{\max} = 0.896 \times 1200 = 1075\ A$$

As in Table C.3, the presented loss factor taking into account skin effect allow 1.32% increase only compared to the h^2-rule, due to the low harmonics spectrum, while the harmonic loss factor, allow 8.6% increase in the current.

Table C.3—Comparison of the maximum allowed non-sinusoidal load current for different harmonic loss factors F_{HL}

Harmonic factor	F_{HL}	$I_{\max}$ (A)
$h^2\left(\frac{I_h}{I}\right)^2$	3.129	1062
$\left(h^2 F\left(\frac{\xi_h}{\xi_R}\right)\right)\left(\frac{I_h}{I}\right)^2$	2.875	1075
$\frac{h\xi_h}{F\xi_R}\left(\frac{I_h}{I}\right)^2$	1.683	1157

C.4 Conclusion

The assumption that the eddy current loss in transformer windings is proportional to the square of the frequency is reasonable for transformers with small conductors and subject to low harmonics. For a combination of large conductors and high harmonics such assumption leads to a conservative result. It predicts losses which are higher than the actual case due to the ignored penetration depth. The actual flux impinging on the conductor will be less due to skin effect.

A corrected harmonic loss factor which takes into account the depth of penetration can be applied to determine more accurately the capability of transformers subject to non-sinusoidal load currents.

Annex D

(informative)

Comparison of UL K-factor definition and IEEE Std C57.110 harmonic loss factor definition

D.1 UL definition of K-factor

The definition of the K-factor rating for dry-type transformers is given in UL 1561-2011 [B52] and UL 1562-2013 [B54]. UL defines the K-factor as follows:[20]

a) K-factor: A rating optionally applied to a transformer indicating its suitability for use with loads that draw nonsinusoidal currents.

b) The K-factor equals:

$$\sum_{h=1}^{\infty} I_h^2 \left(\text{pu}\right) h^2$$

where

$I_{\mathrm{h}}\left(\text{pu}\right)$ is the rms current at harmonic h (per unit of rated rms load current)
h is the harmonic order

c) K-factor rated transformers have not been evaluated for use with harmonic loads where the rms current of any singular harmonic greater than the tenth harmonic is greater than 1/h of the fundamental rms current.

D.2 Relationship between K-factor and harmonic loss factor

The UL definition of the K-factor is based on using the transformer rated current in the calculation of per-unit current in the above equation. Substituting the rated current into the UL equation for the K-factor produces Equation (D.1):

$$\text{K-factor} = \sum_{h=1}^{\infty} \left[\frac{I_h}{I_{\mathrm{R}}}\right]^2 h^2 = \frac{1}{I_{\mathrm{R}}^2} \sum_{h=1}^{\infty} I_h^{\ 2}\, h^2 \qquad \text{(D.1)}$$

where

h is the harmonic order
I_h is the rms current at harmonic h (amperes)
I_{R} is the rms fundamental current under rated frequency and rated load conditions (amperes)

The harmonic loss factor, as defined by this standard, is given by Equation (12) as follows:

$$F_{\mathrm{HL}} = \frac{\displaystyle\sum_{h=1}^{h\,=\,h_{\max}} \left[\frac{I_h}{I_1}\right]^2 h^2}{\displaystyle\sum_{h=1}^{h\,=\,h_{\max}} \left[\frac{I_h}{I_1}\right]^2}$$

[20]UL 1561-2011 [B53] and UL 1562-2013 [B54] should be consulted for a complete description of UL requirements for K-factor rated dry-type transformers.

where

F_{HL} is the harmonic loss factor for winding eddy currents
h is the harmonic order
h_{max} is the highest significant harmonic number
I_h is the rms current at harmonic h (amperes)
I_1 is the rms fundamental load current (amperes)

I_1 is a constant and may be moved in front of the summation sign and eliminated as shown in Equation (D.2):

$$F_{HL} = \frac{\dfrac{1}{I_1^2}\displaystyle\sum_{h=1}^{h=h_{max}} I_h^2\, h^2}{\dfrac{1}{I_1^2}\displaystyle\sum_{h=1}^{h=h_{max}} I_h^2} = \frac{\displaystyle\sum_{h=1}^{h=h_{max}} I_h^2\, h^2}{\displaystyle\sum_{h=1}^{h=h_{max}} I_h^2} \tag{D.2}$$

Then rearranging Equation (D.2) produces Equation (D.3):

$$\sum_{h=1}^{h=h_{max}} I_h^2\, h^2 = F_{HL} \sum_{h=1}^{h=h_{max}} I_h^2 \tag{D.3}$$

Substituting Equation (D.3) into Equation (D.1) produces Equation (D.4):

$$\text{K-factor} = \left[\frac{\displaystyle\sum_{h=1}^{h=h_{max}} I_h^2}{I_R^2}\right] F_{HL} \tag{D.4}$$

Equation (D.4) gives the relationship of the harmonic loss factor to the UL K-factor. The harmonic loss factor is a function of the harmonic current spectrum and is independent of the relative magnitude. The UL K-factor is dependent on both the magnitude and the distribution of the harmonic current. For measurements of harmonic currents in existing installations, the numerical value of the K-factor is different from the numerical value of the harmonic loss factor. For a set of harmonic load current measurements, the calculation of the UL K-factor is dependent on the transformer-rated secondary current. For a new transformer with harmonic currents specified as per unit of the rated transformer secondary current, the K-factor and harmonic loss factor have the same numerical values. The numerical value of the K-factor equals the numerical value of the harmonic loss factor only when the square root of the sum of the harmonic currents squared equals the rated secondary current of the transformer.

D.3 Example calculations

Assume an existing installation with a 2500 kVA, 480 V three-phase dry-type transformer. The rated full load rms current on the 480 V winding, or I_R is equal to 3007 A. Harmonic load current measurements were made as given in Table D.1. The K-factor is calculated as shown in Table D.1.

Table D.1—Example of tabulated calculation of the K-factor

h	h^2	I_h	$\frac{I_h}{I_R}$	$\left(\frac{I_h}{I_R}\right)^2$	$\left(\frac{I_h}{I_R}\right)^2 h^2$
1	1	1764	0.58663	0.34414	0.34414
5	25	309	0.58663	0.010560	0.26399

Table continues

Table D.1—Example of tabulated calculation of the K-factor *(continued)*

h	h^2	I_h	$\frac{I_h}{I_R}$	$\left(\frac{I_h}{I_R}\right)^2$	$\left(\frac{I_h}{I_R}\right)^2 h^2$
7	49	195	0.10276	0.0042054	0.20606
11	121	79.4	0.064849	0.00069723	0.084364
13	169	50.5	0.026405	0.00028204	0.047665
17	289	27.1	0.016794	0.000081222	0.023473
19	361	17.7	0.0090123	0.000034648	0.012508
Σ					0.9822
Σ = K-factor = 0.982					

Similar calculations may be made for other transformer kVA ratings applied to the same set of harmonic load current measurements as shown in Table D.1. For transformers rated less than 1500 kVA, the rms value of the harmonic load currents exceeds the rated transformer current. The results are summarized below for the same set of harmonic current measurements. The harmonic loss factor is calculated as shown in 4.6.

Table D.2—Comparison of K-factor and harmonic loss factor for different loads

kVA	Rated current (I_R) (A)	K-factor	Harmonic loss factor (F_{HL})
1500	1804	2.73	2.73
2000	2406	1.53	2.73
2500	3007	0.982	2.73

Annex E

(informative)

Temperature rise testing procedures

E.1 Preferred method of performing a temperature rise test[21]

Solid-state ac power supplies are available over a wide kVA range and are capable of testing most transformers, from small distribution to medium power. As the preferred method of test, these power sources are capable of supplying full current with variable voltage and variable, programmable frequency. This flexible frequency feature is of particular interest for transformers expected to carry harmonic loads. These supplies allow the direct simulation of various harmonic load profiles, providing an accurate means of determining the transformer thermal characteristics.

E.2 Alternative simulated load temperature rise testing procedures[22]

The purpose of the test is to establish the top-liquid temperature rise in steady-state condition, with dissipation of total loss equal to the loss at the nonsinusoidal load current and rated sinusoidal transformer voltage. The test should also establish the average winding temperature rise above liquid under the same conditions for liquid immersed transformers and the average winding temperature rise above ambient for dry-type transformers.

E.2.1 Test method

The test method used may be the loading-back method, the impedance kVA method, or the short-circuit (separate load loss and excitation test) method in accordance with IEEE Std C57.12.90 or IEEE Std C57.12.91 provided the load is adjusted to compensate for harmonic losses.

E.2.2 Dry-type transformers

The basic premise of the simulated test is to determine the temperature rises after inducing losses equivalent to the losses generated by nonsinusoidal load current. These harmonic losses are simulated by increasing the test current by an appropriate amount according to the methods highlighted in E.2.2.1 and E.2.2.2.

E.2.2.1 Load loss simulation—Method I

As an alternative to the actual loading method described in E.2.1, a less accurate simulation may be performed. This simulation requires less equipment and should be used with caution, since it is possible to overload one winding significantly. As such, the procedure is most suitable for small units where the winding eddy losses are similar for both the high-voltage and the low-voltage windings.

The load losses supplied by the transformer under test ($P_{\text{LL-T}}$) are monitored and maintained during the test. These losses are determined as follows in Equation (E.1):

$$P_{\text{LL}-\text{T}} = P_{\text{DC}} \times \left(1 + F_{\text{HL}} \times P_{\text{EC}}\right) \times T_{\text{C}} \tag{E.1}$$

[21] Additional information and specific test procedures are available from a variety of power supply manufacturers.
[22] This document gives simplified calculation methods that can give reasonable estimates for loading. Implicit within this simplified method is the premise that for the same harmonic loss factor, the same losses and hence temperature rise will be produced. This is not necessarily true. Harmonic loading of transformers gives rise to higher losses, and their distribution within the transformer is very different from standard 60 Hz losses, and their magnitudes and location are dependent on complex calculations of geometry and flux penetration depths. These loading methods, therefore, do not give a good simulation of the loss spatial distribution, and hence, they should not be used as definitive methods for the determination of hot-spot temperatures in windings and core or structural components.

where

$P_{\mathrm{LL-T}}$ is the load losses supplied by the transformer under test (watts)
P_{DC} is the total calculated I^2R losses at ambient temperature (watts)
F_{HL} is the harmonic loss factor for winding eddy currents
P_{EC} is the winding eddy-current loss (watts)
T_{C} is the temperature correction factor

$$T_{\mathrm{C}} = \frac{T_{\mathrm{s}} + T_{\mathrm{k}}}{T_{\mathrm{a}} + T_{\mathrm{k}}}$$

where

T_{C} is the temperature correction factor
T_{s} is the maximum acceptable insulation system temperature rise plus 20 °C
T_{k} is the ambient temperature at which the impedance losses and the I^2R losses were determined
T_{a} is 234.5 for copper, or is 225 for EC grade aluminum

(The appropriate values for other grades or alloys may be used.)

P_{EC} (pu) is equal to the assumed eddy-current losses under rated conditions in per unit of rated load I^2R loss and is calculated as follows:

—For transformers rated 300 kVA or less:

$$P_{\mathrm{EC}}(\mathrm{pu}) = \frac{P_{\mathrm{AC}} - P_{\mathrm{DC}}}{P_{\mathrm{DC}} \times T_{\mathrm{C}}^2}$$

—For transformers rated more than 300 kVA:

$$P_{\mathrm{EC}}(\mathrm{pu}) = \frac{C \times \left(P_{\mathrm{AC}} - P_{\mathrm{DC}}\right)}{P_{\mathrm{2-DC}} \times T_{\mathrm{C}}^2}$$

where

$P_{\mathrm{EC}}(\mathrm{pu})$ is the per-unit winding eddy-current loss
P_{AC} is the measured impedance losses at ambient temperature (watts)
P_{DC} is the total calculated I^2R losses at ambient temperature (watts)
T_{C} is the temperature correction factor
$P_{\mathrm{2-DC}}$ is the I^2R losses for the inner winding with the winding at ambient temperature
C is 0.7 for transformers having a turns ratio greater than 4:1 and having one or more windings with a current rating greater than 1000 A
is 0.6 for all other transformers

The impedance losses and the I^2R losses should be determined in accordance with IEEE Std C57.12.90 or IEEE Std C57.12.91.

E.2.2.2 Load loss simulation—Method II

This method is similar to Method I, except it attempts to account for the different eddy-current losses for each winding by establishing an equivalent harmonic current. It may therefore be more appropriate for larger transformers and for units having large differences in the high-voltage and low-voltage winding eddy losses. This simulation also requires less equipment than the actual loading method, but it requires several mathematical corrections.

The load current supplied by the transformer under test may be determined from Equation (17) shown below. The maximum designed current rating is defined as follows:

$$I_{\text{max}}(\text{pu}) = \sqrt{\frac{P_{\text{LL-R}}(\text{pu})}{1 + F_{\text{HL}} \times P_{\text{EC-R}}(\text{pu})}}$$

where

$I_{\text{max}}(\text{pu})$ is the max permissible rms nonsinusoidal load current under rated condition
$P_{\text{LL-R}}(\text{pu})$ is the per-unit load loss under rated conditions
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{\text{EC-R}}(\text{pu})$ is the per-unit winding eddy-current loss under rated conditions

The per-unit test current required to simulate the losses for the harmonic loading in each winding, referred to the rated current, is given as follows in Equation (E.2):

$$I_{\text{T}}(\text{pu}) = \sqrt{\frac{1 + F_{\text{HL}} \times P_{\text{EC-R}}(\text{pu})}{P_{\text{LL-R}}(\text{pu})}} \tag{E.2}$$

where

$I_{\text{T}}(\text{pu})$ is the per-unit rms test current
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{\text{EC-R}}(\text{pu})$ is the per-unit winding eddy-current loss under rated conditions
$P_{\text{LL-R}}(\text{pu})$ is the per-unit load loss under rated conditions

Since the per-unit eddy-current losses in each winding are normally not the same, the following intermediate test factors α_1 and α_2 are defined for convenience for the HV and LV, respectively, as shown in Equation (E.3) and Equation (E.4):

$$\alpha_1 = \sqrt{\frac{1 + F_{\text{HL}} \times P_{\text{EC-1-R}}(\text{pu})}{P_{\text{LL-1-R}}(\text{pu})}} = \sqrt{1 + F_{\text{HL}} \times P_{\text{EC-1-R}}(\text{pu})} \tag{E.3}$$

and

$$\alpha_2 = \sqrt{\frac{1 + F_{\text{HL}} \times P_{\text{EC-2-R}}(\text{pu})}{P_{\text{LL-2-R}}(\text{pu})}} = \sqrt{1 + F_{\text{HL}} \times P_{\text{EC-2-R}}(\text{pu})} \tag{E.4}$$

where

a_1 is the HV test factor
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{\text{EC-1-R}}(\text{pu})$ is the per-unit HV winding eddy-current loss under rated conditions
$P_{\text{LL-1-R}}(\text{pu})$ is the per-unit load loss under rated conditions for the HV winding
a_2 is the LV test factor
$P_{\text{EC-2-R}}(\text{pu})$ is the per-unit LV winding eddy-current loss under rated conditions
$P_{\text{LL-2-R}}(\text{pu})$ is the per-unit load loss under rated conditions for the LV winding

The current for each winding may then be calculated as follows in Equation (E.5) and Equation (E.6):

$$I_{1-\mathrm{T}} = \alpha_1 \times I_{1-\mathrm{R}} \tag{E.5}$$

and

$$I_{2-\mathrm{T}} = \alpha_2 \times I_{2-\mathrm{R}} \tag{E.6}$$

where

$I_{1-\mathrm{T}}$ is the rms HV test current (amperes)
α_1 is the HV test factor
$I_{1-\mathrm{R}}$ is the rms HV current under rated conditions (amperes)
$I_{2-\mathrm{T}}$ is the rms LV test current (amperes)
α_2 is the LV test factor
$I_{2-\mathrm{R}}$ is the rms LV current under rated conditions (amperes)

Since the current of only one winding may determine the value of the test current, an intermediate value is established as a compromise, as shown in Equation (E.7) and Equation (E.8):

$$\alpha = \frac{\alpha_1 + \alpha_2}{2} \tag{E.7}$$

where

α is the test factor
α_1 is the HV test factor
α_2 is the LV test factor

$$I_{\mathrm{T}}(\mathrm{pu}) = \alpha \times I_{\mathrm{R}}(\mathrm{pu}) \tag{E.8}$$

where

$I_{\mathrm{T}}(\mathrm{pu})$ is the per-unit test current
α is the test factor
$I_{\mathrm{R}}(\mathrm{pu})$ is the per-unit current under rated conditions

The individual temperature rise values determined from testing with this compromise load current are then corrected by Equation (E.9) and Equation (E.10):

$$\theta_1 = \theta_{1-\mathrm{T}} \times \left(\tfrac{\alpha_1}{\alpha}\right)^2 \tag{E.9}$$

and

$$\theta_2 = \theta_{2-\mathrm{T}} \times \left(\tfrac{\alpha_2}{\alpha}\right)^2 \tag{E.10}$$

where

θ_1 is the calculated average HV winding temperature rise (°C)

θ_2 is the calculated average LV winding temperature rise (°C)
$\theta_{1-\mathrm{T}}$ is the measured average HV winding temperature rise (°C)
$\theta_{2-\mathrm{T}}$ is the measured average LV winding temperature rise (°C)
α is the test factor
α_1 is the HV test factor
α_2 is the LV test factor

Note that this procedure overloads one winding during the temperature test. If the winding to be overloaded is not capable of withstanding the expected temperature, a lower value of α may be used. This value should not be less than the lower value of α_1 and α_2.

E.2.3 Liquid immersed transformers

The top-liquid rise must first be determined for liquid immersed transformers, before the winding temperature rises may be established. This requires the injection of the total losses, which is composed of the load loss plus the no load loss. The load loss is the total loss developed from the non-sinusoidal load current and includes the winding dc losses, winding eddy losses, and other stray losses. The relationship of these losses is defined by Equation (19):

$$P_{\mathrm{LL}} = P + F_{\mathrm{HL}} \times P_{\mathrm{EC-R}} + F_{\mathrm{HL-STR}} \times P_{\mathrm{OSL-R}}$$

where

P_{LL} is the load loss (watts)
P is the I^2R loss portion of the load loss (watts)
F_{HL} is the harmonic loss factor for winding eddy currents
$P_{\mathrm{EC-R}}$ is the winding eddy-current loss under rated conditions (watts)
$F_{\mathrm{HL-STR}}$ is the harmonic loss factor for other stray losses
$P_{\mathrm{OSL-R}}$ is the other stray loss under rated conditions (watts)

The no-load loss corresponds to the rated transformer voltage. The total injected losses are then measured, and the fundamental power-frequency current is adjusted to give the specified test value of the total loss.

When the top-liquid temperature rise has been established, the test continues with a sinusoidal test current equivalent to the total load loss (P_{LL}). This condition is maintained for 1 h during which measurements of liquid and cooling medium temperatures are made. The equivalent test current is determined by Load Loss Simulation Method II, as discussed in E.2.2.2. At the end of the temperature rise test, the temperatures of the two windings are determined according to standard methods in the references noted in E.2.1.

Annex F

(informative)

Derivation of the ratio of highest winding eddy loss to average

F.1 Introduction

A high percentage of the leakage flux flowing axially in and between the windings is attracted radially inward at the ends of the windings because there is a lower reluctance return path through the core leg than through the unit permeability space outside the windings. In the absence of other information, the inner winding may be assumed to be the low-voltage winding. As a result, the highest magnitude of the radial component of leakage flux density (and highest eddy loss) occurs in the end regions of the inner winding and in the layers adjacent to the main gap.

F.2 Wire wound windings

For wire wound windings, the ratio of the highest winding eddy loss to the average winding eddy loss is a function of the number of layers in the winding. This ratio for a two winding transformer may be derived as follows:

The classical solution of the eddy-current loss component, neglecting the reaction of these currents on the leakage field, may be written for all conductors [B3], [B34]:

$$L_e = (\beta H_i)^4 \left(\frac{(5m^2 - 1)}{45} \right) \tag{F.1}$$

where:

- β is a constant for a given coil structure and material
- H_i is the conductor height
- m is the number of winding layers

The solution for the conductor q is written as follows:

$$L_e = (\beta H_i)^4 \left(\frac{(q^2 - q)}{3} + \frac{4}{45} \right) \tag{F.2}$$

The highest loss is typically in the last layer of the inner winding. Setting m equal to q, the ratio of the maximum loss to the average loss may be found by dividing Equation 2 by Equation 1, resulting in the following equation:

$$L_{e-\max} = \frac{15(m^2 - m) + 4}{(5m^2 - 1)} \tag{F.3}$$

Table F.1—Ratio of maximum to average winding eddy loss

Number of layers	Ratio highest to average winding eddy loss
1	1.00
2	1.79

Table continues

Table F.1—Ratio of maximum to average winding eddy loss *(continued)*

Number of layers	Ratio highest to average winding eddy loss
3	2.14
4	2.33
6	2.54
8	2.67
12	2.76
20	2.85

The maximum ratio possible asymptotically approaches 3. The maximum eddy loss in the hottest spot region for a wire winding is therefore 300% of the average winding eddy loss.

F.3 Foil wound windings

For foil wound windings, the ratio of the highest winding eddy loss to the average winding eddy loss can be significantly higher than the 300% calculated as a maximum for the wire windings and is much more difficult to calculate. The losses are generally determined by finite element analysis and are then verified by test. However, testing indicates that the actual heating generated by these eddy losses tends to be moderated by the very high conductivity of the foil conductor [B38].

F.4 Conservative loss ratio estimate

An estimate of the ratio of the highest winding eddy loss to the average winding eddy loss is required in order to calculate the heating effects on a given transformer. However, this maximum is a function of many factors including the conductor type used in the windings. Three hundred percent is the maximum ratio for a wire winding, but determining an effective estimate for a foil winding is more complicated. Since this document is intended to provide a conservative estimate rather than a precise value, a ratio of 400% has been historically used from the first published version of this document. This ratio provides a compromise value that can be safely used for estimating purposes when the type of winding is unknown. If a more precise calculation is necessary, the manufacturer should be contacted to assist in determining the heating effects of the given harmonic loading.

Annex G

(informative)

Sample transformer loss data

Table G.1 is manufacturing data calculated for a number of liquid immersed specialty type distribution transformers. Units are all 50 Hz.

Table G.1—Calculated winding eddy and other stray losses in 50 Hz distribution transformers

kVA	HV (V)	LV (V)	ILV (A)	LV cond	HV cond	Eddy LV (W)	Eddy HV (W)	Stray (W)	Eddy %	Stray %
350	6300	725	279	Foil	Round	76	56	82	62	38
550	6300	725	438	Foil	Round	97	78	106	62	38
630	4160	480	758	Foil	Round	99	73	153	53	57
630	4160	480	758	Foil	Rectangular	93	96	188	43	57
750	6300	725	597	Foil	Round	60	64	136	48	52
1000	13 800	725	796	Foil	Round	135	117	251	50	50
1125	6300	725	896	Foil	Rectangular	159	180	251	57	43
1300	13 800	2305	326	Foil	Round	115	106	116	66	34
1500	6300	725	1195	Foil	Rectangular	149	383	374	59	41
1973	6000	1903	299+299	Foil	Rectangular	321	278	686	47	53
2125	6300	725	1692	Foil	Rectangular	235	430	798	45	55
2750	13 800	720	2205	Foil	Rectangular	398	416	1398	37	63
2966	13 800	2305	743	Foil	Rectangular	239	510	506	60	40
3450	6300	725	2747	Foil	Rectangular	623	1577	2880	43	57

Figure G.1 and Figure G.2 show loss curves for a large sample of 7500 solid cast transformers over the period of 2011 to 2015. The units are three phase, 50 Hz and 60 Hz with high voltages ranging from 500 to 36.6 kV. The capacity range is 630 kVA to 20 MVA.

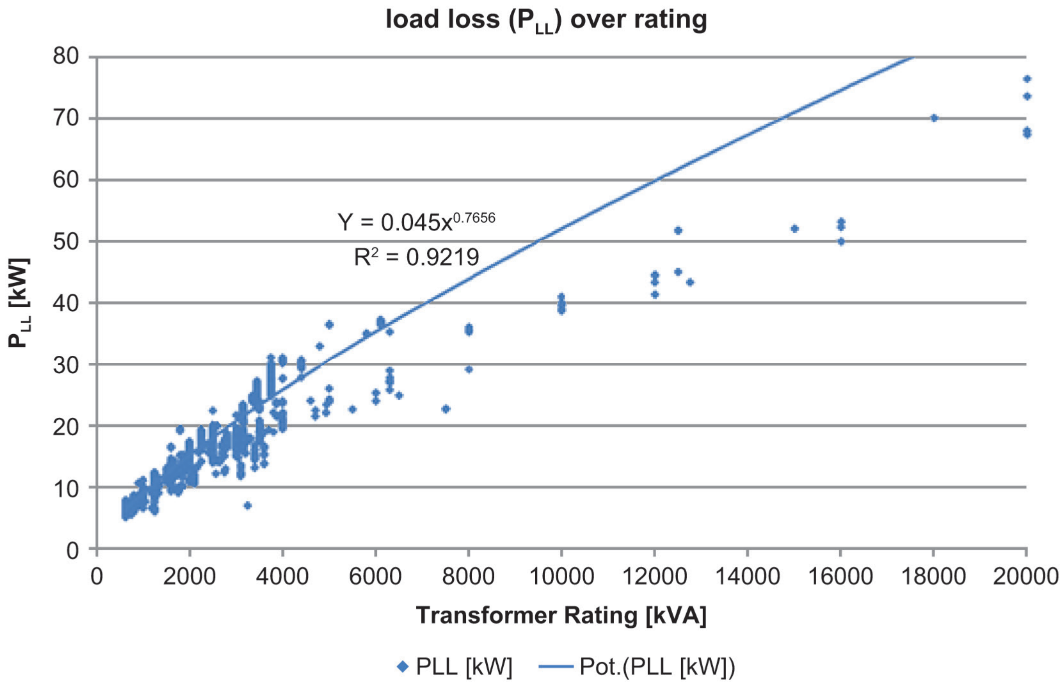

Figure G.1—Load losses (Pk) versus kVA rating for solid cast transformers

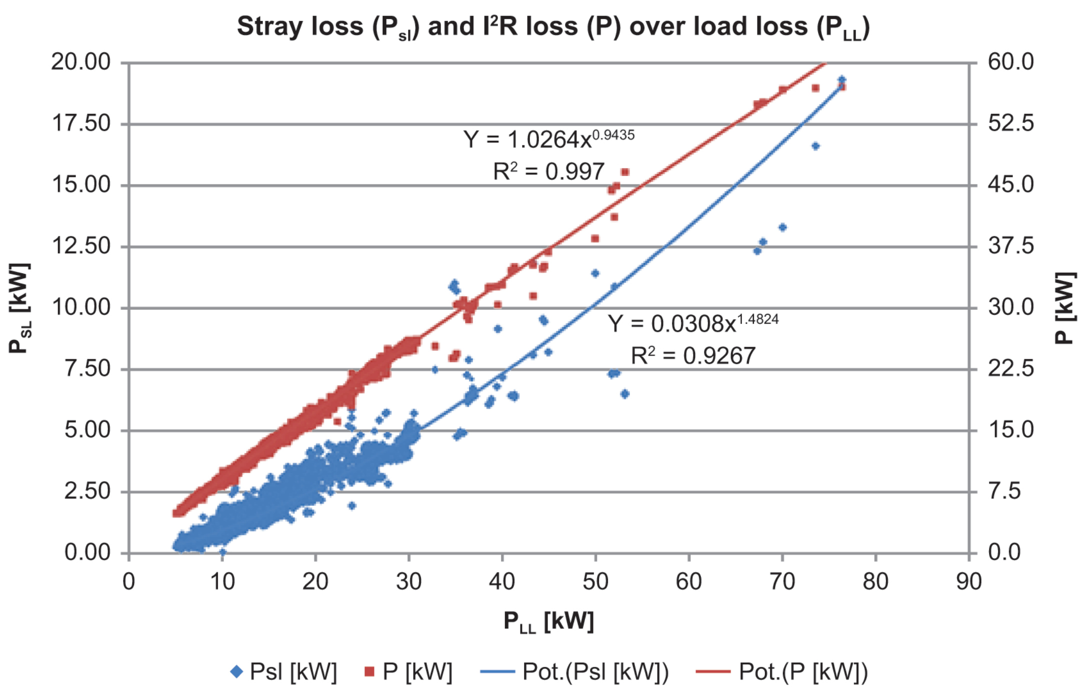

Figure G.2—I^2R losses and other stray losses (Pz) versus load losses (Pk) for solid cast transformers

Table G.2 is manufacturing data calculated for a number of more standard liquid immersed distribution transformers. Units are all 60 Hz.

Table G.1—Calculated winding eddy and other stray losses in 60 Hz distribution transformers

75 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	220	197	32.7	Rect Cu	Round Cu	563	9	584	2	37	1195	48	11	22.9	81.8	18.2
13 200	220	197	103.9	Rect Cu	Round Cu	452	4	308	1	34	799	39	5	12.8	80.0	20.0
23 100	220	197	181.9	Rect Cu	Round Al	575	10	741	0	38	1364	48	10	20.8	100.0	0.0
34 500	220	197	271.6	Rect Cu	Round Cu	519	7	638	0	40	1204	47	7	14.9	100.0	0.0
75 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	220	197	32.8	Foil Cu	Round Cu	540	25	629	2	16	1212	43	27	62.8	92.6	7.4
13 800	220	197	108.7	Foil Cu	Round Cu	447	25	662	0	16	1150	41	25	61.0	100.0	0.0
23 100	220	197	181.9	Foil Cu	Round Cu	668	40	927	0	16	1651	56	40	71.4	100.0	0.0
34 500	220	197	271.6	Foil Cu	Round Cu	560	24	730	0	17	1331	41	24	58.5	100.0	0.0
150 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	220	394	32.8	Rect Cu	Round Cu	632	24	1191	5	100	1952	129	29	22.5	82.8	17.2
13 200	220	394	103.9	Rect Cu	Round Cu	777	9	545	4	90	1425	103	13	12.6	69.2	30.8
23 100	220	394	181.9	Rect Cu	Round Cu	1034	7	1149	1	103	2294	111	8	7.2	87.5	12.5
34 500	220	394	271.6	Rect Cu	Round Cu	755	14	1281	1	111	2162	126	15	11.9	93.3	6.7

Table continues

Table G.1—Calculated winding eddy and other stray losses in 60 Hz distribution transformers *(continued)*

75 kVA	**60 Hz**															
150 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	220	394	32.7	Foil Cu	Round Cu	1065	67	736	12	42	1922	121	79	65.3	84.8	15.2
13 200	220	394	103.9	Foil Cu	Round Cu	492	18	725	2	29	1266	49	20	40.8	90.0	10.0
23 100	220	394	181.9	Foil Cu	Round Cu	1030	84	1011	1	51	2177	136	85	62.5	98.8	1.2
34 500	220	394	271.7	Foil Cu	Round Cu	975	77	1095	1	50	2198	128	78	60.9	98.7	1.3
300 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
2400	480	361	8.7	Rect Cu	Round Cu	1640	486	3420	42	155	5743	683	528	77.3	92.0	8.0
13 800	220	787	108.6	Rect Cu	Round Cu	1283	46	1883	13	318	3543	377	59	15.6	78.0	22.0
13 800	220	787	108.6	Rect Cu	Round Cu	1824	125	2565	6	309	4829	440	131	29.8	95.4	4.6
23 100	220	787	181.9	Rect Cu	Round Cu	1545	27	2016	7	333	3928	367	34	9.3	79.4	20.6
34 500	220	787	271.6	Rect Cu	Round Cu	1438	24	1942	4	340	3748	368	28	7.6	85.7	14.3
300 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	220	787	32.8	Foil Cu	Round Cu	1524	146	1721	35	157	3583	338	181	53.6	80.7	19.3
13 800	220	787	108.7	Foil Cu	Round Cu	1649	121	1670	13	156	3609	290	134	46.2	90.3	9.7
13 800	220	787	108.6	Foil Cu	Round Cu	2161	133	2118	10	154	4576	297	143	48.1	93.0	7.0

Table continues

Table G.1—Calculated winding eddy and other stray losses in 60 Hz distribution transformers *(continued)*

75 kVA	60 Hz															
13 800	380	456	62.9	Foil Cu	Round Cu	2107	115	2162	9	95	4488	219	124	56.6	92.7	7.3
23 100	380	456	105.3	Foil Al	Round Cu	2165	166	2941	4	103	5379	273	170	62.3	97.6	2.4
34 500	220	787	271.6	Foil Cu	Round Cu	1525	113	1910	3	171	3722	287	116	40.4	97.4	2.6
500 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	380	760	19.9	Foil Al	Round Cu	2479	220	3184	58	169	6110	447	278	62.2	79.1	20.9
13 800	380	760	62.9	Foil Al	Round Cu	2169	341	2794	21	216	5541	578	362	62.6	94.2	5.8
24 200	380	760	110.3	Foil Al	Round Cu	3240	186	3925	6	183	7540	375	192	51.2	96.9	3.1
34 500	380	760	157.2	Foil Al	Round Cu	2154	177	3891	4	191	6417	372	181	48.7	97.8	2.2
500 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	220	1312	32.7	Foil Al	Round Cu	2315	219	3216	52	357	6159	628	271	43.2	80.8	19.2
13 800	220	1312	108.6	Foil Al	Round Cu	3755	144	3944	18	346	8207	508	162	31.9	88.9	11.1
24 200	220	1312	190.5	Foil Al	Round Cu	3548	184	4066	7	363	8168	554	191	34.5	96.3	3.7
34 500	220	1312	271.6	Foil Al	Round Cu	3548	195	4175	5	371	8294	571	200	35.0	97.5	2.5
1000 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
6300	230	2510	47.4	Foil Al	Rect Cu	5604	199	4164	80	1104	11 151	1383	279	20.2	71.3	28.7
11 000	400	1443	47.6	Foil Al	Rect Cu	4931	365	4480	83	428	10 287	876	448	51.1	81.5	18.5
13 800	380	1519	62.9	Foil Al	Round Cu	4255	552	5534	70	662	11 073	1284	622	48.4	88.7	11.3
23 100	380	1519	105.3	Foil Cu	Round Cu	5865	221	5504	68	639	12 297	928	289	31.1	76.5	23.5

Table continues

Table G.1—Calculated winding eddy and other stray losses in 60 Hz distribution transformers *(continued)*

75 kVA	60 Hz															
23 100	440	1312	91.0	Foil Cu	Round Cu	6118	226	7368	54	549	14 315	829	280	33.8	80.7	19.3
24 200	380	1519	110.3	Foil Al	Round Al	4809	245	4589	69	541	10 253	855	314	36.7	78.0	22.0
30 000	400	1443	129.9	Foil Al	Round Cu	4697	429	4626	82	502	10 336	1013	511	50.4	84.0	16.0
34 500	380	1519	157.3	Foil Al	Round Al	4809	271	5304	40	565	10 989	876	311	35.5	87.1	12.9
2000 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	400	2887	18.0	Foil Al	Rect Cu	12 093	406	15 399	186	1755	29 839	2347	592	25.2	68.6	31.4
6300	460	2510	23.7	Foil Al	Rect Cu	6214	679	8660	383	1528	17 464	2590	1062	41.0	63.9	36.1
13 200	400	2887	57.1	Foil Al	Layer Rect Al	9182	812	10 977	296	1879	23 146	2987	1108	37.1	73.3	26.7
13 800	440	2624	54.3	Foil Cu	Rect Cu	8605	494	9792	333	2143	21 367	2970	827	27.8	59.7	40.3
34 500	400	2887	149.4	Foil Al	Rect Cu	8428	541	6229	166	2564	17 928	3271	707	21.6	76.5	23.5
3000 kVA	**60 Hz**															
HV (V)	**LV (V)**	**I LV (A)**	**Turns ratio**	**LV**	**HV**	**I^2R LV (W)**	**Eddy LV (W)**	**I^2R HV (W)**	**Eddy HV (W)**	**Stray (W)**	**Total (W)**	**Stray+ Eddy (W)**	**Eddy (W)**	**Eddy %**	**Eddy LV %**	**Eddy HV %**
4160	400	4330	18.0	Foil Al	Rect Cu	15 540	691	18 352	392	6015	40 990	7098	1083	15.3	63.8	36.2
6300	400	4330	27.3	Foil Al	Rect Cu	13 065	766	15 285	498	4410	34 024	5674	1264	22.3	60.6	39.4
13 800	380	4558	62.9	Foil Al	Rect Cu	13 569	1037	14 402	394	6865	36 267	8296	1431	17.2	72.5	27.5
13 800	440	3937	54.3	Foil Al	Rect Cu	11 936	1173	14 868	437	5029	33 443	6639	1610	24.3	72.9	27.1
22 000	690	2510	55.3	Foil Cu	Rect Cu	10 786	578	12 492	237	1748	25 841	2563	815	31.8	70.9	29.1
34 500	460	3765	129.9	Foil Cu	Rect Cu	9484	672	9337	487	4919	24 899	6078	1159	19.1	58.0	42.0

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